AI & COMPUTING

The Mathematicall Praeface to Elements of Geometrie of Euclid of Megara

John Dee

[Transcriber’s Note:

This e-text includes characters that will only display in UTF-8 (Unicode) text readers, including a few words of Greek:

Τακτικὴ [Taktikê] ã ẽ õ ũ [overline or tilde to show following -n or -m] □ □ [leaf symbol; pointing-finger symbol] ‡ [double-ended dagger, used in size notations (below)]

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Some aspects of the original book had to be modified for all versions of this plain-text file.

Superscript letters are shown with ^: y^e, y^t.

Marginal quotation marks are shown inline as “ and ”, approximating the beginning and end of the marked passage. In the original text, no quotation marks were printed inline.

Paragraphs are broken up for sidenotes, with blank lines before and after. Original paragraph breaks are shown as two blank lines. Brackets within the body text are in the original.

All sidenotes except the one beginning “This noble Earle” were printed in italics; markup has been omitted to reduce visual clutter.

At least four sizes of text were used, often in combination with _italics_. The variants are shown here as:

+‡very large‡+ +larger+ =smaller=

Further errors and anomalies are listed at the end of the text, along with those Euclid citations identified by number.]

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THE ELEMENTS OF GEOMETRIE

of the most auncient Philosopher _EVCLIDE_ of Megara.

_Faithfully (now first) translated into the Englishe toung, by _H. Billingsley_, Citizen of London_.

Whereunto are annexed certaine Scholies, Annotations, and Inuentions, of the best Mathematiciens, both of time past, and in this our age.

_With a very fruitfull Præface made by _M. I. Dee_, specifying the chiefe Mathematicall Sciẽces, what they are, and wherunto commodious: where, also, are disclosed certaine new Secrets Mathematicall and Mechanicall, vntill these our daies, greatly missed._

Imprinted at London by _Iohn Daye_.

 The Translator to the Reader.

_There is (gentle Reader) nothing (the word of God onely set apart) which so much beautifieth and adorneth the soule and minde of mã, as doth the knowledge of good artes and sciences: as the knowledge of naturall and morall Philosophie. The one setteth before our eyes, the creatures of God, both in the heauens aboue, and in the earth beneath: in which as in a glasse, we beholde the exceding maiestie and wisedome of God, in adorning and beautifying them as we see: in geuing vnto them such wonderfull and manifolde proprieties, and naturall workinges, and that so diuersly and in such varietie: farther in maintaining and conseruing them continually, whereby to praise and adore him, as by S. Paule we are taught. The other teacheth vs rules and preceptes of vertue, how, in common life amongest men, we ought to walke vprightly: what dueties pertaine to our selues, what pertaine to the gouernment or good order both of an housholde, and also of a citie or common wealth. The reading likewise of histories, conduceth not a litle, to the adorning of the soule & minde of man, a studie of all men cõmended: by it are seene and knowen the artes and doinges of infinite wise men gone before vs. In histories are contained infinite examples of heroicall vertues to be of vs followed, and horrible examples of vices to be of vs eschewed. Many other artes also there are which beautifie the minde of man: but of all other none do more garnishe & beautifie it, then those artes which are called Mathematicall. Unto the knowledge of which no man can attaine, without the perfecte knowledge and instruction of the principles, groundes, and Elementes of Geometrie. But perfectly to be instructed in them, requireth diligent studie and reading of olde auncient authors. Amongest which, none for a beginner is to be preferred before the most auncient Philosopher _Euclide_ of _Megara_. For of all others he hath in a true methode and iuste order, gathered together whatsoeuer any before him had of these Elementes written: inuenting also and adding many thinges of his owne: wherby he hath in due forme accomplished the arte: first geuing definitions, principles, & groundes, wherof he deduceth his Propositions or conclusions, in such wonderfull wise, that that which goeth before, is of necessitie required to the proufe of that which followeth. So that without the diligent studie of _Euclides_ Elementes, it is impossible to attaine vnto the perfecte knowledge of Geometrie, and consequently of any of the other Mathematicall sciences. Wherefore considering the want & lacke of such good authors hitherto in our Englishe tounge, lamenting also the negligence, and lacke of zeale to their countrey in those of our nation, to whom God hath geuen both knowledge, & also abilitie to translate into our tounge, and to publishe abroad such good authors, and bookes (the chiefe instrumentes of all learninges): seing moreouer that many good wittes both of gentlemen and of others of all degrees, much desirous and studious of these artes, and seeking for them as much as they can, sparing no paines, and yet frustrate of their intent, by no meanes attaining to that which they seeke: I haue for their sakes, with some charge & great trauaile, faithfully translated into our vulgare toũge, & set abroad in Print, this booke of _Euclide_. Whereunto I haue added easie and plaine declarations and examples by figures, of the definitions. In which booke also ye shall in due place finde manifolde additions, Scholies, Annotations, and Inuentions: which I

haue gathered out of many of the most famous & chiefe Mathematiciẽs, both of old time, and in our age: as by diligent reading it in course, ye shall well perceaue. The fruite and gaine which I require for these my paines and trauaile, shall be nothing els, but onely that thou gentle reader, will gratefully accept the same: and that thou mayest thereby receaue some profite: and moreouer to excite and stirre vp others learned, to do the like, & to take paines in that behalfe. By meanes wherof, our Englishe tounge shall no lesse be enriched with good Authors, then are other straunge tounges: as the Dutch, French, Italian, and Spanishe: in which are red all good authors in a maner, found amongest the Grekes or Latines. Which is the chiefest cause, that amongest thẽ do florishe so many cunning and skilfull men, in the inuentions of straunge and wonderfull thinges, as in these our daies we see there do. Which fruite and gaine if I attaine vnto, it shall encourage me hereafter, in such like sort to translate, and set abroad some other good authors, both pertaining to religion (as partly I haue already done) and also pertaining to the Mathematicall Artes. Thus gentle reader farewell._ (?¿)

[Decoration]

 TO THE VNFAINED LOVERS of truthe, and constant Studentes of Noble _Sciences, _IOHN DEE_ of London, hartily_ wisheth grace from heauen, and most prosperous _successe in all their honest attemptes and_ exercises.

Diuine _Plato_, the great Master of many worthy Philosophers, and the constant auoucher, and pithy perswader of _Vnum_, _Bonum_, and _Ens_: in his Schole and Academie, sundry times (besides his ordinary Scholers) was visited of a certaine kinde of men, allured by the noble fame of _Plato_, and the great commendation of hys profound and profitable doctrine. But when such Hearers, after long harkening to him, perceaued, that the drift of his discourses issued out, to conclude, this _Vnum_, _Bonum_, and _Ens_, to be Spirituall, Infinite, Æternall, Omnipotent, &c. Nothyng beyng alledged or expressed, How, worldly goods: how, worldly dignitie: how, health, Strẽgth or lustines of body: nor yet the meanes, how a merueilous sensible and bodyly blysse and felicitie hereafter, might be atteyned: Straightway, the fantasies of those hearers, were dampt: their opinion of _Plato_, was clene chaunged: yea his doctrine was by them despised: and his schole, no more of them visited. Which thing, his Scholer, _Aristotle_, narrowly cõsidering, founde the cause therof, to be, “For that they had no forwarnyng and information, in generall,” whereto his doctrine tended. For, so, might they haue had occasion, either to haue forborne his schole hauntyng: (if they, then, had misliked his Scope and purpose) or constantly to haue continued therin: to their full satisfaction: if such his finall scope & intent, had ben to their desire. Wherfore, _Aristotle_, euer, after that, vsed in brief, to forewarne his owne Scholers and hearers, “both of what matter, and also to what ende, he tooke in hand to speake, or teach.” While I consider the diuerse trades of these two excellent Philosophers (and am most sure, both, that _Plato_ right well, otherwise could teach: and that _Aristotle_ mought boldely, with his hearers, haue dealt in like sorte as _Plato_ did) I am in no little pang of perplexitie: Bycause, that, which I mislike, is most easy for me to performe (and to haue _Plato_ for my exãple.) And that, which I know to be most commendable: and (in this first bringyng, into common handling, the

Artes Mathematicall) to be most necessary: is full of great difficultie and sundry daungers. Yet, neither do I think it mete, for so straunge matter (as now is ment to be published) and to so straunge an audience, to be bluntly, at first, put forth, without a peculiar Preface: Nor (Imitatyng _Aristotle_) well can I hope, that accordyng to the amplenes and dignitie of the _State Mathematicall_, I am able, either playnly to prescribe the materiall boundes: or precisely to expresse the chief purposes, and most wonderfull applications therof. And though I am sure, that such as did shrinke from _Plato_ his schole, after they had perceiued his finall conclusion, would in these thinges haue ben his most diligent hearers (so infinitely mought their desires, in fine and at length, by our _Artes Mathematicall_ be satisfied) yet, by this my Præface & forewarnyng, Aswell all such, may (to their great behofe) the soner, hither be allured: as also the _Pythagoricall_, and _Platonicall_ perfect scholer, and the constant profound Philosopher, with more ease and spede, may (like the Bee,) gather, hereby, both wax and hony.

[The intent of this Preface.]

Wherfore, seyng I finde great occasion (for the causes alleged, and farder, in respect of my _Art Mathematike generall_) to vse “a certaine forewarnyng and Præface, whose content shalbe, that mighty, most plesaunt, and frutefull _Mathematicall Tree_, with his chief armes and second (grifted) braunches: Both, what euery one is, and also, what commodity, in generall, is to be looked for, aswell of griff as stocke: And forasmuch as this enterprise is so great, that, to this our tyme, it neuer was (to my knowledge) by any achieued: And also it is most hard, in these our drery dayes, to such rare and straunge Artes, to wyn due and common credit:” Neuertheles, if, for my sincere endeuour to satisfie your honest expectation, you will but lend me your thãkefull mynde a while: and, to such matter as, for this time, my penne (with spede) is hable to deliuer, apply your eye or eare attentifely: perchaunce, at once, and for the first salutyng, this Preface you will finde a lesson long enough. And either you will, for a second (by this) be made much the apter: or shortly become, well hable your selues, of the lyons claw, to coniecture his royall symmetrie, and farder propertie. Now then, gentle, my frendes, and countrey men, Turne your eyes, and bend your myndes to that doctrine, which for our present purpose, my simple talent is hable to yeld you.

All thinges which are, & haue beyng, are found vnder a triple diuersitie generall. For, either, they are demed Supernaturall, Naturall, or, of a third being. Thinges Supernaturall, are immateriall, simple, indiuisible, incorruptible, & vnchangeable. Things Naturall, are materiall, compounded, diuisible, corruptible, and chaungeable. Thinges Supernaturall, are, of the minde onely, comprehended: Things Naturall, of the sense exterior, ar hable to be perceiued. In thinges Naturall, probabilitie and coniecture hath place: But in things Supernaturall, chief demõstration, & most sure Science is to be had. By which properties & comparasons of these two, more easily may be described, the state, condition, nature and property of those thinges, which, we before termed of a third being: which, by a peculier name also, are called _Thynges Mathematicall_. For, these, beyng (in a maner) middle, betwene thinges supernaturall and naturall: are not so absolute and excellent, as thinges

supernatural: Nor yet so base and grosse, as things naturall: But are thinges immateriall: and neuerthelesse, by materiall things hable somewhat to be signified. And though their particular Images, by Art, are aggregable and diuisible: yet the generall _Formes_, notwithstandyng, are constant, vnchaungeable, vntrãsformable, and incorruptible. Neither of the sense, can they, at any tyme, be perceiued or iudged. Nor yet, for all that, in the royall mynde of man, first conceiued. But, surmountyng the imperfectiõ of coniecture, weenyng and opinion: and commyng short of high intellectuall cõceptiõ, are the Mercurial fruite of _Dianœticall_ discourse, in perfect imagination subsistyng. A meruaylous newtralitie haue these thinges _Mathematicall_, and also a straunge participatiõ betwene thinges supernaturall, immortall, intellectual, simple and indiuisible: and thynges naturall, mortall, sensible, compounded and diuisible. Probabilitie and sensible prose, may well serue in thinges naturall: and is commendable: In Mathematicall reasoninges, a probable Argument, is nothyng regarded: nor yet the testimony of sense, any whit credited: But onely a perfect demonstration, of truthes certaine, necessary, and inuincible: vniuersally and necessaryly concluded: is allowed as sufficient for “an Argument exactly and purely Mathematical.”

[Note the worde, Vnit, to expresse the Greke Monas, & not Vnitie: as we haue all, commonly, till now, vsed.]

Of _Mathematicall_ thinges, are two principall kindes: namely, _Number_, and _Magnitude_.

[Number.]

Number, we define, to be, a certayne Mathematicall Sũme, of _Vnits_. And, an _Vnit_, is that thing Mathematicall, Indiuisible, by participation of some likenes of whose property, any thing, which is in deede, or is counted One, may resonably be called One. We account an _Vnit_, a thing _Mathematicall_, though it be no Number, and also indiuisible: because, of it, materially, Number doth consist: which, principally, is a thing _Mathematicall_.

[Magnitude.]

Magnitude is a thing _Mathematicall_, by participation of some likenes of whose nature, any thing is iudged long, broade, or thicke. “A thicke _Magnitude_ we call a _Solide_, or a _Body_. What _Magnitude_ so euer, is Solide or Thicke, is also broade, & long. A broade magnitude, we call a _Superficies_ or a Plaine. Euery playne magnitude, hath also length. A long magnitude, we terme a _Line_. A _Line_ is neither thicke nor broade, but onely long: Euery certayne Line, hath two endes:

[A point.]

The endes of a line, are _Pointes_ called. A _Point_, is a thing _Mathematicall_, indiuisible, which may haue a certayne determined situation.” If a Poynt moue from a determined situation, the way wherein it moued, is also a _Line_: mathematically produced, whereupon, of the auncient Mathematiciens,

[A Line.]

a _Line_ is called the race or course of a _Point_. A Poynt we define, by the name of a thing Mathematicall: though it be no Magnitude, and indiuisible: because it is the propre ende, and bound of a Line: which is a true _Magnitude_.

[Magnitude.]

And _Magnitude_ we may define to be that thing _Mathematicall_, which is diuisible for euer, in partes diuisible, long, broade or thicke. Therefore though a Poynt be no _Magnitude_, yet _Terminatiuely_, we recken it a thing _Mathematicall_ (as I sayd) by reason it is properly the end, and bound of a line. Neither _Number_, nor _Magnitude_, haue any Materialitie. First, we will consider of _Number_, and of the Science _Mathematicall_, to it appropriate, called _Arithmetike_: and afterward of _Magnitude_, and his Science, called _Geometrie_. But that name contenteth me not: whereof a word or two hereafter shall be sayd. How Immateriall and free from all matter, _Number_ is, who doth not perceaue? yea, who doth not wonderfully wõder at it? For, neither pure _Element_, nor _Aristoteles, Quinta Essentia_, is hable to serue for Number, as his propre matter. Nor yet the puritie and simplenes of Substance Spirituall or Angelicall, will be found propre enough thereto. And therefore the great & godly Philosopher _Anitius Boetius_, sayd: _Omnia quæcun[que] a primæua rerum natura constructa sunt, Numerorum videntur ratione formata. Hoc enim fuit principale in animo Conditoris Exemplar_. That is: +_All thinges (which from the very first originall being of thinges, haue bene framed and made) do appeare to be Formed by the reason of Numbers. For this was the principall example or patterne in the minde of the Creator_.+ O comfortable allurement, O rauishing perswasion, to deale with a Science, whose Subiect, is so Auncient, so pure, so excellent, so surmounting all creatures, so vsed of the Almighty and incomprehensible wisdome of the Creator, in the distinct creation of all creatures: in all their distinct partes, properties, natures, and vertues, by order, and most absolute number, brought, from _Nothing_, to the _Formalitie_ of their being and state. By _Numbers_ propertie therefore, of vs, by all possible meanes, (to the perfection of the Science) learned, we may both winde and draw our selues into the inward and deepe search and vew, of all creatures distinct vertues, natures, properties, and _Formes_: And also, farder, arise, clime, ascend, and mount vp (with Speculatiue winges) in spirit, to behold in the Glas of Creation, the _Forme of Formes_, the _Exemplar Number_ of all thinges _Numerable_: both visible and inuisible, mortall and immortall, Corporall and Spirituall. Part of this profound and diuine Science, had _Ioachim_ the Prophesier atteyned vnto: by _Numbers Formall, Naturall_, and _Rationall_, forseyng, concludyng, and forshewyng great particular euents, long before their comming. His bookes yet remainyng, hereof, are good profe: And the noble Earle of _Mirandula_, (besides that,) a sufficient witnesse: that _Ioachim, in his prophesies, proceded by no other way, then by Numbers Formall_. And this Earle hym selfe, in Rome,

[Ano. 1488.]

* set vp 900. Conclusions, in all kinde of Sciences, openly to be disputed of: and among the rest, in his Conclusions _Mathematicall_, (in the eleuenth Conclusion) hath in Latin, this English sentence. _By Numbers, a way is had, to the searchyng out, and vnderstandyng of euery thyng, hable to be knowen. For the verifying of which

Conclusion, I promise to aunswere to the 74. Questions, vnder written, by the way of Numbers_. Which Cõclusions, I omit here to rehearse: aswell auoidyng superfluous prolixitie: as, bycause _Ioannes Picus, workes_, are commonly had. But, in any case, I would wish that those Conclusions were red diligently, and perceiued of such, as are earnest Obseruers and Considerers of the constant law of nũbers: which is planted in thyngs Naturall and Supernaturall: and is prescribed to all Creatures, inuiolably to be kept. For, so, besides many other thinges, in those Conclusions to be marked, it would apeare, how sincerely, & within my boundes, I disclose the wonderfull mysteries, by numbers, to be atteyned vnto.

Of my former wordes, easy it is to be gathered, that _Number_ hath a treble state: One, in the Creator: an other in euery Creature (in respect of his complete constitution:) and the third, in Spirituall and Angelicall Myndes, and in the Soule of mã. In the first and third state, _Number_, is termed _Number Numbryng_. But in all Creatures, otherwise, _Number_, is termed _Nũber Numbred_. And in our Soule, Nũber beareth such a swaye, and hath such an affinitie therwith: that some of the old _Philosophers_ taught, _Mans Soule, to be a Number mouyng it selfe_. And in dede, in vs, though it be a very Accident: yet such an Accident it is, that before all Creatures it had perfect beyng, in the Creator, Sempiternally. _Number Numbryng_ therfore, is the discretion discerning, and distincting of thinges. But in God the Creator, This discretion, in the beginnyng, produced orderly and distinctly all thinges. For his _Numbryng_, then, was his Creatyng of all thinges. And his Continuall _Numbryng_, of all thinges, is the Conseruation of them in being: And, where and when he will lacke an _Vnit_: there and then, that particular thyng shalbe _Discreated_. Here I stay. But our Seuerallyng, distinctyng, and _Numbryng_, createth nothyng: but of Multitude considered, maketh certaine and distinct determination. And albeit these thynges be waighty and truthes of great importance, yet (by the infinite goodnes of the Almighty _Ternarie_,) Artificiall Methods and easy wayes are made, by which the zelous Philosopher, may wyn nere this Riuerish _Ida_, this Mountayne of Contemplation: and more then Contemplation. And also, though _Number_, be a thyng so Immateriall, so diuine, and æternall: yet by degrees, by litle and litle, stretchyng forth, and applying some likenes of it, as first, to thinges Spirituall: and then, bryngyng it lower, to thynges sensibly perceiued: as of a momentanye sounde iterated: then to the least thynges that may be seen, numerable: And at length, (most grossely,) to a multitude of any corporall thynges seen, or felt: and so, of these grosse and sensible thynges, we are trayned to learne a certaine Image or likenes of numbers: and to vse Arte in them to our pleasure and proffit. So grosse is our conuersation, and dull is our apprehension: while mortall Sense, in vs, ruleth the common wealth of our litle world. Hereby we say, Three Lyons, are three: or a _Ternarie_. Three Egles, are three, or a _Ternarie_.

[□]

Which * _Ternaries_, are eche, the _Vnion_, _knot_, and _Vniformitie_, of three discrete and distinct _Vnits_. That is, we may in eche _Ternarie_, thrise, seuerally pointe, and shew a part, _One_, _One_, and _One_. Where, in Numbryng, we say One, two, Three. But how farre, these visible Ones, do differre from our Indiuisible Vnits

(in pure _Arithmetike_, principally considered) no man is ignorant. Yet from these grosse and materiall thynges, may we be led vpward, by degrees, so, informyng our rude Imagination, toward the cõceiuyng of _Numbers_, absolutely (:Not supposing, nor admixtyng any thyng created, Corporall or Spirituall, to support, conteyne, or represent those _Numbers_ imagined:) that at length, we may be hable, to finde the number of our owne name, gloriously exemplified and registred in the booke of the _Trinitie_ most blessed and æternall.

But farder vnderstand, that vulgar Practisers, haue Numbers, otherwise, in sundry Considerations: and extend their name farder, then to Numbers, whose least part is an _Vnit_. For the common Logist, Reckenmaster, or Arithmeticien, in hys vsing of Numbers: of an Vnit, imagineth lesse partes: and calleth them _Fractions_. As of an _Vnit_, he maketh an halfe, and thus noteth it, ½. and so of other, (infinitely diuerse) partes of an _Vnit_. Yea and farder, hath, _Fractions of Fractions. &c_. And, forasmuch, as, _Addition_, _Substraction_, _Multiplication_, _Diuision_ and _Extraction of Rotes_, are the chief, and sufficient partes of _Arithmetike_:

[Arithmetike.]

which is, the _Science that demonstrateth the properties, of Numbers, and all operatiõs, in numbers to be performed_:

[Note.]

“How often, therfore, these fiue sundry sortes of Operations, do, for the most part, of their execution, differre from the fiue operations of like generall property and name, in our Whole numbers practisable, So often, (for a more distinct doctrine) we, vulgarly account and name it, an other kynde of _Arithmetike_.” And by this reason:

[1.]

the Consideration, doctrine, and working, in whole numbers onely: where, of an _Vnit_, is no lesse part to be allowed: is named (as it were) an _Arithmetike_ by it selfe. And so of the _Arithmetike of Fractions_.

[2.]

In lyke sorte, the necessary, wonderfull and Secret doctrine of Proportion, and proportionalytie hath purchased vnto it selfe a peculier maner of handlyng and workyng: and so may seme an other forme of _Arithmetike_.

[3.]

Moreouer, the _Astronomers_, for spede and more commodious calculation, haue deuised a peculier maner of orderyng nũbers, about theyr circular motions, by Sexagenes, and Sexagesmes. By Signes, Degrees and Minutes &c. which commonly is called the _Arithmetike_ of _Astronomical_ or _Phisicall Fractions_. That, haue I briefly noted, by the name of _Arithmetike Circular_. Bycause it is also vsed in circles, not _Astronomicall. &c._

[4.]

Practise hath led _Numbers_ farder, and hath framed them, to take vpon them, the shew of _Magnitudes_ propertie: Which is _Incommensurabilitie_ and _Irrationalitie_. (For in pure _Arithmetike_, an _Vnit_, is the common Measure of all Numbers.) And, here, Nũbers are become, as Lynes, Playnes and Solides: some tymes _Rationall_, some tymes _Irrationall_. And haue propre and peculier characters, (as $^{2}\surd$. $^{3}\surd$. and so of other. Which is to signifie _Rote Square, Rote Cubik: and so forth_:) & propre and peculier fashions in the fiue principall partes: Wherfore the practiser, estemeth this, a diuerse _Arithmetike_ from the other. Practise bryngeth in, here, diuerse compoundyng of Numbers: as some tyme, two, three, foure (or more) _Radicall_ nũbers, diuersly knit, by signes, of More & Lesse: as thus $^{2}\surd 12 + {}^{3}\surd 15$. Or thus $^{4}\surd 19 + {}^{3}\surd 12 - {}^{2}\surd 2$. &c. And some tyme with whole numbers, or fractions of whole Number, amõg them: as $20 + {}^{2}\surd 24$. $^{3}\surd 16 + 33 - {}^{2}\surd 10$. $^{4}\surd 44 + 12¼ + {}^{3}\surd 9$. And so, infinitely, may hap the varietie. After this: Both the one and the other hath fractions incident: and so is this _Arithmetike_ greately enlarged, by diuerse exhibityng and vse of Compositions and mixtynges. Consider how, I (beyng desirous to deliuer the student from error and Cauillation) do giue to this _Practise_, the name of the _Arithmetike of Radicall numbers_: Not, of _Irrationall_ or _Surd Numbers_: which other while, are Rationall: though they haue the Signe of a Rote before them, which, _Arithmetike_ of whole Numbers most vsuall, would say they had no such Roote: and so account them _Surd Numbers_: which, generally spokẽ, is vntrue: as _Euclides_ tenth booke may teach you. Therfore to call them, generally, _Radicall Numbers_, (by reason of the signe $\surd$. prefixed,) is a sure way: and a sufficient generall distinction from all other ordryng and vsing of Numbers: And yet (beside all this) Consider: the infinite desire of knowledge, and incredible power of mans Search and Capacitye: how, they, ioyntly haue waded farder (by mixtyng of speculation and practise) and haue found out, and atteyned to the very chief perfection (almost) of _Numbers_ Practicall vse. Which thing, is well to be perceiued in that great Arithmeticall Arte of _Æquation_: commonly called the _Rule of Coss._ or _Algebra_. The Latines termed it, _Regulam Rei & Census_, that is, the +_Rule of the thyng and his value_+. With an apt name: comprehendyng the first and last pointes of the worke. And the vulgar names, both in Italian, Frenche and Spanish, depend (in namyng it,) vpon the signification of the Latin word, _Res_: +_A thing_+: vnleast they vse the name of _Algebra_. And therin (commonly) is a dubble error. The one, of them, which thinke it to be of _Geber_ his inuentyng: the other of such as call it _Algebra_. For, first, though _Geber_ for his great skill in Numbers, Geometry, Astronomy, and other maruailous Artes, mought haue semed hable to haue first deuised the sayd Rule: and also the name carryeth with it a very nere likenes of _Geber_ his name: yet true it is, that a _Greke_ Philosopher and Mathematicien, named _Diophantus_, before _Geber_ his tyme, wrote 13. bookes therof (of which, six are yet extant: and I had them to *vse,

[* Anno. 1550.]

of the famous Mathematicien, and my great frende, _Petrus Montaureus_:) And secondly, the very name, is _Algiebar_, and not _Algebra_: as by the Arabien _Auicen_, may be proued: who hath these precise wordes in Latine, by _Andreas Alpagus_ (most perfect in the Arabik tung) so translated. _Scientia faciendi Algiebar & Almachabel. i. Scientia inueniendi numerum ignotum, per additionem Numeri, &

diuisionem & æquationem_. Which is to say: +_The Science of workyng Algiebar and Almachabel_+, that is, the +_Science of findyng an vnknowen number, by Addyng of a Number, & Diuision & æquation_+. Here haue you the name: and also the principall partes of the Rule, touched. To name it, _The rule, or Art of Æquation_, doth signifie the middle part and the State of the Rule. This Rule, hath his peculier Characters:

[5.]

and the principal partes of _Arithmetike_, to it appertayning, do differre from the other _Arithmeticall operations_. This _Arithmetike, hath Nũbers_ Simple, Cõpound, Mixt: and Fractions, accordingly. This Rule, and _Arithmetike of Algiebar_, is so profound, so generall and so (in maner) conteyneth the whole power of Numbers Application practicall: that mans witt, can deale with nothyng, more proffitable about numbers: nor match, with a thyng, more mete for the diuine force of the Soule, (in humane Studies, affaires, or exercises) to be tryed in. Perchaunce you looked for, (long ere now,) to haue had some particular profe, or euident testimony of the vse, proffit and Commodity of Arithmetike vulgar, in the Common lyfe and trade of men. Therto, then, I will now frame my selfe: But herein great care I haue, least length of sundry profes, might make you deme, that either I did misdoute your zelous mynde to vertues schole: or els mistrust your hable witts, by some, to gesse much more. A profe then, foure, fiue, or six, such, will I bryng, as any reasonable man, therwith may be persuaded, to loue & honor, yea learne and exercise the excellent Science of _Arithmetike_.

And first: who, nerer at hand, can be a better witnesse of the frute receiued by _Arithmetike_, then all kynde of Marchants? Though not all, alike, either nede it, or vse it. How could they forbeare the vse and helpe of the Rule, called the Golden Rule? Simple and Compounde: both forward and backward? How might they misse _Arithmeticall_ helpe in the Rules of Felowshyp: either without tyme, or with tyme? and betwene the Marchant & his Factor? The Rules of Bartering in wares onely: or part in wares, and part in money, would they gladly want? Our Marchant venturers, and Trauaylers ouer Sea, how could they order their doynges iustly and without losse, vnleast certaine and generall Rules for Exchaũge of money, and Rechaunge, were, for their vse, deuised? The Rule of Alligation, in how sundry cases, doth it conclude for them, such precise verities, as neither by naturall witt, nor other experience, they, were hable, els, to know? And (with the Marchant then to make an end) how ample & wonderfull is the Rule of False positions? especially as it is now, by two excellent Mathematiciens (of my familier acquayntance in their life time) enlarged? I meane _Gemma Frisius_, and _Simon Iacob_. Who can either in brief conclude, the generall and Capitall Rules? or who can Imagine the Myriades of sundry Cases, and particular examples, in Act and earnest, continually wrought, tried and concluded by the forenamed Rules, onely? How sundry other _Arithmeticall practises_, are commonly in Marchantes handes, and knowledge: They them selues, can, at large, testifie.

The Mintmaster, and Goldsmith, in their Mixture of Metals, either of diuerse kindes, or diuerse values: how are they, or may they, exactly be directed, and meruailously pleasured, if _Arithmetike_ be their guide? And the honorable Phisiciãs, will gladly confesse them selues, much beholding to the Science of _Arithmetike_, and

that sundry wayes: But chiefly in their Art of Graduation, and compounde Medicines. And though _Galenus_, _Auerrois_, _Arnoldus_, _Lullus_, and other haue published their positions, aswell in the quantities of the Degrees aboue Temperament, as in the Rules, concluding the new _Forme_ resulting: yet a more precise, commodious, and easy _Method_, is extant: by a Countreyman of ours

[R. B.]

(aboue 200. yeares ago) inuented. And forasmuch as I am vncertaine, who hath the same: or when that litle Latin treatise, (as the Author writ it,) shall come to be Printed: (Both to declare the desire I haue to pleasure my Countrey, wherin I may: and also, for very good profe of Numbers vse, in this most subtile and frutefull, Philosophicall Conclusion,) I entend in the meane while, most briefly, and with my farder helpe, to communicate the pith therof vnto you.

First describe a circle: whose diameter let be an inch. Diuide the Circumference into foure equall partes. Frõ the Center, by those 4. sections, extend 4. right lines: eche of 4. inches and a halfe long: or of as many as you liste, aboue 4. without the circumference of the circle: So that they shall be of 4. inches long (at the least) without the Circle. Make good euident markes, at euery inches end. If you list, you may subdiuide the inches againe into 10. or 12. smaller partes, equall. At the endes of the lines, write the names of the 4. principall elementall Qualities. _Hote_ and _Colde_, one against the other. And likewise _Moyst_ and _Dry_, one against the other. And in the Circle write _Temperate_. Which _Temperature_ hath a good Latitude: as appeareth by the Complexion of man. And therefore we haue allowed vnto it, the foresayd Circle: and not a point Mathematicall or Physicall.

[* Take some part of Lullus counsayle in his booke de Q. Essentia.]

Now, when you haue two thinges Miscible, whose degrees are * truely knowen: Of necessitie, either they are of one Quantitie and waight, or of diuerse. If they be of one Quantitie and waight: whether their formes, be Contrary Qualities, or of one kinde (but of diuerse intentions and degrees) or a _Temperate_, and a Contrary, _The forme resulting of their Mixture, is in the Middle betwene the degrees of the formes mixt_. As for example, let _A_, be _Moist_ in the first degree: and _B_, _Dry_ in the third degree. Adde 1. and 3. that maketh 4: the halfe or middle of 4. is 2. This 2. is the middle, equally distant from _A_ and _B_

[* Note.]

(for the * _Temperament_ is counted none. And for it, you must put a Ciphre, if at any time, it be in mixture).

HOTE +C | | + | | + | | +E | MOIST A TEMPERATE B DRYE +------+------+------+------+------+------+------+------+ |D | + | | + | | + | | + COLD

Counting then from _B_, 2. degrees, toward _A_: you finde it to be _Dry_ in the first degree: So is the _Forme resulting_ of the Mixture of _A_, and _B_, in our example. I will geue you an other example. Suppose, you haue two thinges, as _C_, and _D_: and of _C_, the Heate to be in the 4. degree: and of _D_, the Colde, to be remisse, euen

vnto the _Temperament_. Now, for _C_, you take 4: and for _D_, you take a Ciphre: which, added vnto 4, yeldeth onely 4. The middle, or halfe, whereof, is 2. Wherefore the _Forme resulting_ of _C_, and _D_, is Hote in the second degree: for, 2. degrees, accounted from _C_, toward _D_, ende iuste in the 2. degree of heate. Of the third maner, I will geue also an example: which let be this:

[Note.]

I haue a liquid Medicine whose Qualitie of heate is in the 4. degree exalted: as was _C_, in the example foregoing: and an other liquid Medicine I haue: whose Qualitie, is heate, in the first degree. Of eche of these, I mixt a like quantitie: Subtract here, the lesse frõ the more: and the residue diuide into two equall partes: whereof, the one part, either added to the lesse, or subtracted from the higher degree, doth produce the degree of the Forme resulting, by this mixture of _C_, and _E_. As, if from 4. ye abate 1. there resteth 3. the halfe of 3. is 1½: Adde to 1. this 1½: you haue 2½. Or subtract from 4. this 1½: you haue likewise 2½ remayning. Which declareth, the _Forme resulting_, to be _Heate_, in the middle of the third degree.

[The Second Rule.]

“But if the Quantities of two thinges Commixt, be diuerse, and the Intensions (of their Formes Miscible) be in diuerse degrees, and heigthes. (Whether those Formes be of one kinde, or of Contrary kindes, or of a Temperate and a Contrary, _What proportion is of the lesse quantitie to the greater, the same shall be of the difference, which is betwene the degree of the Forme resulting, and the degree of the greater quantitie of the thing miscible, to the difference, which is betwene the same degree of the Forme resulting, and the degree of the lesse quantitie_. As for example. Let two pound of Liquor be geuen, hote in the 4. degree: & one pound of Liquor be geuen, hote in the third degree.” I would gladly know the Forme resulting, in the Mixture of these two Liquors. Set downe your nũbers in order, thus. _____ | | | | {P}. _2._ | _Hote. 4._ | | | | | {P}. _1._ | _Hote. 3._ | |_____|_____|

Now by the rule of Algiebar, haue I deuised a very easie, briefe, and generall maner of working in this case. Let vs first, suppose that _Middle Forme resulting_, to be 1{x}: as that Rule teacheth. And because (by our Rule, here geuen) as the waight of 1. is to 2: So is the difference betwene 4. (the degree of the greater quantitie) and 1{x}: to the difference betwene 1{x} and 3: (the degree of the thing, in lesse quãtitie. And with all, 1{x}, being alwayes in a certaine middell, betwene the two heigthes or degrees). For the first difference, I set 4-1{x}: and for the second, I set 1{x}-3. And, now againe, I say, as 1. is to 2. so is 4-1{x} to 1{x}-3. Wherfore, of these foure proportionall numbers, the first and the fourth Multiplied, one by the other, do make as much, as the second and the third Multiplied the one by the other. Let these Multiplications be made accordingly. And of the first and the fourth, we haue 1{x}-3. and of the second & the third, 8-2{x}. Wherfore, our Æquation is betwene 1{x}-3: and 8-2{x}. Which may be reduced, according to the Arte of Algiebar: as, here, adding 3. to eche part, geueth the Æquation, thus, 1{x}=11-2{x}. And yet againe, contracting, or Reducing it: Adde to eche part, 2{x}: Then haue you 3{x} æquall to 11: thus represented 3{x}=11. Wherefore, diuiding 11. by 3: the Quotient is 3⅔: the

Valew of our 1{x}, _Coss_, or _Thing_, first supposed. And that is the heigth, or Intension of the _Forme resulting:_ which is, _Heate_, in two thirdes of the fourth degree: And here I set the shew of the worke in conclusion, thus. The proufe hereof is easie: by subtracting 3. from 3⅔, resteth ⅔. Subtracte the same heigth of the Forme resulting, (which is 3⅔) frõ 4: then resteth ⅓: You see, that ⅔ is double to ⅓: as 2.{P}. is double to 1.{P}. So should it be: by the rule here geuen. Note. As you added to eche part of the Æquation, 3: so if ye first added to eche part 2{x}, it would stand, 3{x}-3=8. And now adding to eche part 3: you haue (as afore) 3{x}=11.

| _____ | | | _ | {P}. _2._ | _Hote. 4._ | ⅓ _ _The forme_ | | | _ _3⅔ resulting._ | {P}. _1._ | _Hote. 3._ | _ ⅔ |
|---|---|---

Mathematicals, that _Ælianus_ (the best writer therof,) in his worke, to the _Emperour Hadrianus_, by his perfection, in the Mathematicals, (beyng greater, then other before him had,) thinketh his booke to passe all other the excellent workes, written of that Art, vnto his dayes. For, of it, had written _Æneas_: _Cyneas_ of _Thessaly_: _Pyrrhus Epirota_: and _Alexander_ his sonne: _Clearchus_: _Pausanias_: _Euangelus_: _Polybius_, familier frende to _Scipio_: _Eupolemus_: _Iphicrates_, _Possidonius_: and very many other worthy Capitaines, Philosophers and Princes of Immortall fame and memory: Whose fayrest floure of their garland (in this feat) was _Arithmetike_: and a litle perceiuerance, in _Geometricall_ Figures. But in many other cases doth _Arithmetike_ stand the Capitaine in great stede. As in proportionyng of vittayles, for the Army, either remaining at a stay: or suddenly to be encreased with a certaine number of Souldiers: and for a certain tyme. Or by good Art to diminish his company, to make the victuals, longer to serue the remanent, & for a certaine determined tyme: if nede so require. And so in sundry his other accountes, Reckeninges, Measurynges, and proportionynges, the wise, expert, and Circumspect Capitaine will affirme the Science of _Arithmetike_, to be one of his chief Counsaylors, directers and aiders. Which thing (by good meanes) was euident to the Noble, the Couragious, the loyall, and Curteous

[□]

Iohn, late Earle of Warwicke. Who was a yong Gentleman, throughly knowne to very few. Albeit his lusty valiantnes, force, and Skill in Chiualrous feates and exercises: his humblenes, and frendelynes to all men, were thinges, openly, of the world perceiued. But what rotes (otherwise,) vertue had fastened in his brest, what Rules of godly and honorable life he had framed to him selfe: what vices, (in some then liuing) notable, he tooke great care to eschew: what manly vertues, in other noble men, (florishing before his eyes,) he Sythingly aspired after: what prowesses he purposed and ment to achieue: with what feats and Artes, he began to furnish and fraught him selfe, for the better seruice of his Kyng and Countrey, both in peace & warre. These (I say) his Heroicall Meditations, forecastinges and determinations, no twayne, (I thinke) beside my selfe, can so perfectly, and truely report. And therfore, in Conscience, I count it my part, for the honor, preferment, & procuring of vertue (thus, briefly) to haue put his Name, in the Register of _Fame Immortall_.

To our purpose. This _Iohn_, by one of his actes (besides many other: both in England and Fraunce, by me, in him noted.) did disclose his harty loue to vertuous Sciences: and his noble intent, to excell in Martiall prowesse: When he, with humble request, and instant Solliciting: got the best Rules (either in time past by Greke or Romaine, or in our time vsed: and new Stratagemes therin deuised) for ordring of all Companies, summes and Numbers of mẽ, (Many, or few) with one kinde of weapon, or mo, appointed: with Artillery, or without: on horsebacke, or on fote: to giue, or take onset: to seem many, being few: to seem few, being many. To marche in battaile or Iornay: with many such feates, to Foughten field, Skarmoush, or Ambushe appartaining:

[This noble Earle, dyed Anno. 1554. skarse of 24. yeares of age: hauing no issue by his wife: Daughter to the Duke of Somerset.]

And of all these, liuely designementes (most curiously) to be in velame parchement described: with Notes & peculier markes, as the Arte requireth: and all these Rules, and descriptions Arithmeticall, inclosed in a riche Case of Gold, he vsed to weare about his necke: as his Iuell most precious, and Counsaylour most trusty. Thus, _Arithmetike_, of him, was shryned in gold: Of _Numbers_ frute, he had good hope. Now, Numbers therfore innumerable, in _Numbers_ prayse, his shryne shall finde.

What nede I, (for farder profe to you) of the Scholemasters of Iustice, to require testimony: how nedefull, how frutefull, how skillfull a thing _Arithmetike_ is? I meane, the Lawyers of all sortes. Vndoubtedly, the Ciuilians, can meruaylously declare: how, neither the Auncient Romaine lawes, without good knowledge of _Numbers art_, can be perceiued: Nor (Iustice in infinite Cases) without due proportion, (narrowly considered,) is hable to be executed. How Iustly, & with great knowledge of Arte, did _Papinianus_ institute a law of partition, and allowance, betwene man and wife after a diuorce? But how _Accursius_, _Baldus_, _Bartolus_, _Iason_, _Alexander_, and finally _Alciatus_, (being otherwise, notably well learned) do iumble, gesse, and erre, from the æquity, art and Intent of the lawmaker: _Arithmetike_ can detect, and conuince: and clerely, make the truth to shine. Good _Bartolus_, tyred in the examining & proportioning of the matter: and with _Accursius_ Glosse, much cumbred: burst out, and sayd: _Nulla est in toto libro, hac glossa difficilior: Cuius computationem nec Scholastici nec Doctores intelligunt. &c._ That is: +_In the whole booke, there is no Glosse harder then this: Whose accoumpt or reckenyng, neither the Scholers, nor the Doctours vnderstand. &c._+ What can they say of _Iulianus_ law, _Si ita Scriptum. &c._ Of the Testators will iustly performing, betwene the wife, Sonne and daughter? How can they perceiue the æquitie of _Aphricanus_, _Arithmeticall_ Reckening, where he treateth of _Lex Falcidia_? How can they deliuer him, from his Reprouers: and their maintainers: as _Ioannes_, _Accursius Hypolitus_ and _Alciatus_? How Iustly and artificially, was _Africanus_ reckening made? Proportionating to the Sommes bequeathed, the Contributions of eche part? Namely, for the hundred presently receiued, 17-1/7. And for the hundred, receiued after ten monethes, 12-6/7: which make the 30: which were to be cõtributed by the legataries to the heire. For, what proportion, 100 hath to 75: the same hath 17-1/7 to 12-6/7: Which is Sesquitertia: that is, as 4, to 3. which make 7. Wonderfull many places, in the Ciuile law, require an expert _Arithmeticien_, to vnderstand the deepe Iudgemẽt, & Iust determinatiõ of the Auncient Romaine Lawmakers. But much more expert ought he to be, who should be hable, to decide with æquitie, the infinite varietie of Cases, which do, or may happen, vnder euery one of those lawes and ordinances Ciuile. Hereby, easely, ye may now coniecture: that in the Canon law: and in the lawes of the Realme (which with vs, beare the chief Authoritie), Iustice and equity might be greately preferred, and skilfully executed, through due skill of Arithmetike, and proportions appertainyng. The worthy Philosophers, and prudent lawmakers (who haue written many bookes _De Republica:_ How the best state of Common wealthes might be procured and mainteined,) haue very well determined of Iustice: (which, not onely, is the Base and foundacion of Common weales: but also the totall perfection of all our workes, words, and thoughtes:) defining it,

[Iustice.]

“to be that vertue, by which, to euery one, is rendred, that to him appertaineth.” God challengeth this at our handes, to be honored as God: to be loued, as a father: to be feared as a Lord & master. Our neighbours proportiõ, is also prescribed of the Almighty lawmaker: which is, to do to other, euen as we would be done vnto. These proportions, are in Iustice necessary: in duety, commendable: and of Common wealthes, the life, strength, stay and florishing. _Aristotle_ in his _Ethikes_ (to fatch the sede of Iustice, and light of direction, to vse and execute the same) was fayne to fly to the perfection, and power of Numbers: for proportions Arithmeticall and Geometricall. _Plato_ in his booke called _Epinomis_ (which boke, is the Threasury of all his doctrine) where, his purpose is, to seke a Science, which, when a man had it, perfectly: he might seme, and so be, in dede, _Wise_. He, briefly, of other Sciences discoursing, findeth them, not hable to bring it to passe: But of the Science of Numbers, he sayth. _Illa, quæ numerum mortalium generi dedit, id profecto efficiet. Deum autem aliquem, magis quam fortunam, ad salutem nostram, hoc munus nobis arbitror contulisse. &c. Nam ipsum bonorum omnium Authorem, cur non maximi boni, Prudentiæ dico, causam arbitramur? +That Science, verely, which hath taught mankynde number, shall be able to bryng it to passe. And, I thinke, a certaine God, rather then fortune, to haue giuen vs this gift, for our blisse. For, why should we not Iudge him, who is the Author of all good things, to be also the cause of the greatest good thyng, namely, Wisedome?+_ There, at length, he proueth _Wisedome_ to be atteyned, by good Skill of _Numbers_. With which great Testimony, and the manifold profes, and reasons, before expressed, you may be sufficiently and fully persuaded: of the perfect Science of _Arithmetike_, to make this accounte: That

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of all Sciences, next to _Theologie_, it is most diuine, most pure, most ample and generall, most profounde, most subtile, most commodious and most necessary. Whose next Sister, is the Absolute Science of _Magnitudes_: of which (by the Direction and aide of him, whose _Magnitude_ is Infinite, and of vs Incomprehensible) I now entend, so to write, that both with the _Multitude_, and also with the _Magnitude_ of Meruaylous and frutefull verities, you (my frendes and Countreymen) may be stird vp, and awaked, to behold what certaine Artes and Sciences, (to our vnspeakable behofe) our heauenly father, hath for vs prepared, and reuealed, by sundry _Philosophers_ and _Mathematiciens_.

Both, _Number_ and _Magnitude_, haue a certaine Originall sede, (as it were) of an incredible property: and of man, neuer hable, Fully, to be declared. Of _Number_, an Vnit, and of _Magnitude_, a Poynte, doo seeme to be much like Originall causes: But the diuersitie neuerthelesse, is great. We defined an _Vnit_, to be a thing Mathematicall Indiuisible: A Point, likewise, we sayd to be a Mathematicall thing Indiuisible. And farder, that a Point may haue a certaine determined Situation: that is, that we may assigne, and prescribe a Point, to be here, there, yonder. &c. Herein, (behold) our Vnit is free, and can abyde no bondage, or to be tyed to any place, or seat: diuisible or indiuisible. Agayne, by reason, a Point may haue a Situation limited to him: a certaine motion, therfore (to a place, and from a place) is to a Point incident and appertainyng. But an _Vnit_, can not be imagined to haue any motion. A Point, by

his motion, produceth, Mathematically, a line: (as we sayd before) which is the first kinde of Magnitudes, and most simple: An _Vnit_, can not produce any number. A Line, though it be produced of a Point moued, yet, it doth not consist of pointes: Number, though it be not produced of an _Vnit_, yet doth it Consist of vnits, as a materiall cause. But formally,

[Number.]

Number, is the Vnion, and Vnitie of Vnits. Which vnyting and knitting, is the workemanship of our minde: which, of distinct and discrete Vnits, maketh a Number: by vniformitie, resulting of a certaine multitude of Vnits. And so, euery number, may haue his least part, giuen: namely, an Vnit: But not of a Magnitude, (no, not of a Lyne,) the least part can be giuẽ: by cause, infinitly, diuision therof, may be conceiued. All Magnitude, is either a Line, a Plaine, or a Solid. Which Line, Plaine, or Solid, of no Sense, can be perceiued, nor exactly by hãd (any way) represented: nor of Nature produced: But, as (by degrees) Number did come to our perceiuerance: So, by visible formes, we are holpen to imagine, what our Line Mathematicall, is. What our Point, is. So precise, are our Magnitudes, that one Line is no broader then an other: for they haue no bredth: Nor our Plaines haue any thicknes. Nor yet our Bodies, any weight: be they neuer so large of dimensiõ. Our Bodyes, we can haue Smaller, then either Arte or Nature can produce any: and Greater also, then all the world can comprehend. Our least Magnitudes, can be diuided into so many partes, as the greatest. As, a Line of an inch long, (with vs) may be diuided into as many partes, as may the diameter of the whole world, from East to West: or any way extended: What priuiledges, aboue all manual Arte, and Natures might, haue our two Sciences Mathematicall? to exhibite, and to deale with thinges of such power, liberty, simplicity, puritie, and perfection? And in them, so certainly, so orderly, so precisely to procede: as, excellent is that workemã Mechanicall Iudged, who nerest can approche to the representing of workes, Mathematically demonstrated?

[□]

And our two Sciences, remaining pure, and absolute, in their proper termes, and in their owne Matter: to haue, and allowe, onely such Demonstrations, as are plaine, certaine, vniuersall, and of an æternall veritye?

[Geometrie.]

This Science of _Magnitude_, his properties, conditions, and appertenances: commonly, now is, and from the beginnyng, hath of all Philosophers, ben called _Geometrie_. But, veryly, with a name to base and scant, for a Science of such dignitie and amplenes. And, perchaunce, that name, by cõmon and secret consent, of all wisemen, hitherto hath ben suffred to remayne: that it might carry with it a perpetuall memorye, of the first and notablest benefite, by that Science, to common people shewed: Which was, when Boundes and meres of land and ground were lost, and confounded (as in _Egypt_, yearely, with the ouerflowyng of _Nilus_, the greatest and longest riuer in the world) or, that ground bequeathed, were to be assigned: or, ground sold, were to be layd out: or (when disorder preuailed) that Commõs were distributed into seueralties. For, where, vpon these & such like occasiõs, Some by

ignorãce, some by negligẽce, Some by fraude, and some by violence, did wrongfully limite, measure, encroach, or challenge (by pretence of iust content, and measure) those landes and groundes: great losse, disquietnes, murder, and warre did (full oft) ensue: Till, by Gods mercy, and mans Industrie, The perfect Science of Lines, Plaines, and Solides (like a diuine Iusticier,) gaue vnto euery man, his owne. The people then, by this art pleasured, and greatly relieued, in their landes iust measuring: & other Philosophers, writing Rules for land measuring: betwene them both, thus, confirmed the name of _Geometria_, that is, (according to the very etimologie of the word) Land measuring. Wherin, the people knew no farder, of Magnitudes vse, but in Plaines: and the Philosophers, of thẽ, had no feet hearers, or Scholers: farder to disclose vnto, then of flat, plaine _Geometrie_. And though, these Philosophers, knew of farder vse, and best vnderstode the etymologye of the worde, yet this name _Geometria_, was of them applyed generally to all sortes of Magnitudes: vnleast, otherwhile, of _Plato_, and _Pythagoras_: When they would precisely declare their owne doctrine. Then, was

[* Plato. 7. de Rep.]

* _Geometria_, with them, _Studium quod circa planum versatur_. But, well you may perceiue by _Euclides Elementes_, that more ample is our Science, then to measure Plaines: and nothyng lesse therin is tought (of purpose) then how to measure Land. An other name, therfore, must nedes be had, for our Mathematicall Science of Magnitudes: which regardeth neither clod, nor turff: neither hill, nor dale: neither earth nor heauen: but is absolute _Megethologia_: not creping on ground, and dasseling the eye, with pole perche, rod or lyne: but “liftyng the hart aboue the heauens, by inuisible lines, and

[□]

immortall beames meteth with the reflexions, of the light incomprehensible: and so procureth Ioye, and perfection vnspeakable.” Of which true vse of our _Megethica_, or _Megethologia_, _Diuine Plato_ seemed to haue good taste, and iudgement: and (by the name of _Geometrie_) so noted it: and warned his Scholers therof: as, in hys seuenth _Dialog_, of the Common wealth, may euidently be sene. Where (in Latin) thus it is: right well translated: _Profecto, nobis hoc non negabunt, Quicun[que] vel paululum quid Geometriæ gustârunt, quin hæc Scientia, contrà, omnino se habeat, quàm de ea loquuntur, qui in ipsa versantur._ In English, thus. +_Verely_+ (sayth _Plato_) +_whosoeuer haue, (but euen very litle) tasted of Geometrie, will not denye vnto vs, this: but that this Science, is of an other condicion, quite contrary to that, which they that are exercised in it, do speake of it._+ And there it followeth, of our _Geometrie_, _Quòd quæritur cognoscendi illius gratia, quod semper est, non & eius quod oritur quando[que] & interit. Geometria, eius quod est semper, Cognitio est. Attollet igitur (ô Generose vir) ad Veritatem, animum: at[que] ita, ad Philosophandum preparabit cogitationem, vt ad supera conuertamus: quæ, nunc, contra quàm decet, ad inferiora deijcimus. &c. Quàm maximè igitur præcipiendum est, vt qui præclarissimam hanc habitãt Civitatem, nullo modo, Geometriam spernant. Nam & quæ præter ipsius propositum, quodam modo esse videntur, haud exigua sunt. &c._ It must nedes be confessed (saith _Plato_) +_That =[Geometrie]= is learned, for the knowyng of that, which is euer: and not of that, which, in tyme, both is bred and is

brought to an ende. &c. Geometrie is the knowledge of that which is euerlastyng. It will lift vp therfore (O Gentle Syr) our mynde to the Veritie: and by that meanes, it will prepare the Thought, to the Philosophicall loue of wisdome: that we may turne or conuert, toward heauenly thinges =[both mynde and thought]= which now, otherwise then becommeth vs, we cast down on base or inferior things. &c. Chiefly, therfore, Commaundement must be giuen, that such as do inhabit this most honorable Citie, by no meanes, despise Geometrie. For euen those thinges =[done by it]= which, in manner, seame to be, beside the purpose of Geometrie: are of no small importance. &c._+ And besides the manifold vses of _Geometrie_, in matters appertainyng to warre, he addeth more, of second vnpurposed frute, and commoditye, arrising by _Geometrie_: saying: _Scimus quin etiam, ad Disciplinas omnes facilius per discendas, interesse omnino, attigerit ne Geometriam aliquis, an non. &c. Hanc ergo Doctrinam, secundo loco discendam Iuuenibus statuamus._ That is. +_But, also, we know, that for the more easy learnyng of all Artes, it importeth much, whether one haue any knowledge in Geometrie, or no. &c. Let vs therfore make an ordinance or decree, that this Science, of young men shall be learned in the second place._+ This was _Diuine Plato_ his Iudgement, both of the purposed, chief, and perfect vse of _Geometrie_: and of his second, dependyng, deriuatiue commodities. And for vs, Christen men, a thousand thousand mo occasions are, to haue nede of the helpe of *

[I. D. * Herein, I would gladly shake of, the earthly name, of Geometrie.]

Megethologicall Contemplations: wherby, to trayne our Imaginations and Myndes, by litle and litle, to forsake and abandon, the grosse and corruptible Obiectes, of our vtward senses: and to apprehend, by sure doctrine demonstratiue, Things Mathematicall. And by them, readily to be holpen and conducted to conceiue, discourse, and conclude of things Intellectual, Spirituall, æternall, and such as concerne our Blisse euerlasting: which, otherwise (without Speciall priuiledge of Illumination, or Reuelation frõ heauen) No mortall mans wyt (naturally) is hable to reach vnto, or to Compasse. And, veryly, by my small Talent (from aboue) I am hable to proue and testifie, that the litterall Text, and order of our diuine Law, Oracles, and Mysteries, require more skill in Numbers, and Magnitudes: then (commonly) the expositors haue vttered: but rather onely (at the most) so warned: & shewed their own want therin. (To name any, is nedeles: and to note the places, is, here, no place: But if I be duely asked, my answere is ready.) And without the litterall, Grammaticall, Mathematicall or Naturall verities of such places, by good and certaine Arte, perceiued, no Spirituall sense (propre to those places, by Absolute _Theologie_) will thereon depend.

[□]

“No man, therfore, can doute, but toward the atteyning of knowledge incomparable, and Heauenly Wisedome: Mathematicall Speculations, both of Numbers and Magnitudes: are meanes, aydes, and guides: ready, certaine, and necessary.” From henceforth, in this my Preface, will I frame my talke, to _Plato_ his fugitiue Scholers: or, rather, to such, who well can, (and also wil,) vse their vtward senses, to the glory of God, the benefite of their Countrey, and their owne secret contentation, or honest preferment, on this earthly Scaffold. To them, I will orderly

recite, describe & declare a great Number of Artes, from our two Mathematicall fountaines, deriued into the fieldes of _Nature_. Wherby, such Sedes, and Rotes, as lye depe hyd in the groũd of _Nature_, are refreshed, quickened, and prouoked to grow, shote vp, floure, and giue frute, infinite, and incredible. And these Artes, shalbe such, as vpon Magnitudes properties do depende, more, then vpon Number. And by good reason we may call them Artes, and Artes Mathematicall Deriuatiue: for (at this tyme) I Define

[An Arte.]

+An Arte, to be a Methodicall cõplete Doctrine, hauing abundancy of sufficient, and peculier matter to deale with, by the allowance of the Metaphisicall Philosopher: the knowledge whereof, to humaine state is necessarye.+ And that I account,

[Art Mathematicall Deriuatiue.]

+An Art Mathematicall deriuatiue, which by Mathematicall demonstratiue Method, in Nũbers, or Magnitudes, ordreth and confirmeth his doctrine, as much & as perfectly, as the matter subiect will admit.+ And for that, I entend to vse the name and propertie of a

[A Mechanitien.]

Mechanicien, otherwise, then (hitherto) it hath ben vsed, I thinke it good, (for distinction sake) to giue you also a brief description, what I meane therby. +A Mechanicien, or a Mechanicall workman is he, whose skill is, without knowledge of Mathematicall demonstration, perfectly to worke and finishe any sensible worke, by the Mathematicien principall or deriuatiue, demonstrated or demonstrable.+ Full well I know, that he which inuenteth, or maketh these demonstrations, is generally called _A speculatiue Mechanicien_: which differreth nothyng from a _Mechanicall Mathematicien_. So, in respect of diuerse actions, one man may haue the name of sundry artes: as, some tyme, of a Logicien, some tymes (in the same matter otherwise handled) of a Rethoricien. Of these trifles, I make, (as now, in respect of my Preface,) small account: to fyle thẽ for the fine handlyng of subtile curious disputers. In other places, they may commaunde me, to giue good reason: and yet, here, I will not be vnreasonable.

[+1.+]

First, then, from the puritie, absolutenes, and Immaterialitie of Principall _Geometrie_, is that kinde of _Geometrie_ deriued, which vulgarly is counted _Geometrie_: and is the +Arte of Measuring sensible magnitudes, their iust quãtities and contentes.+

[Geometrie vulgar.]

This, teacheth to measure, either at hand: and the practiser, to be by the thing Measured: and so, by due applying of Cumpase, Rule, Squire, Yarde, Ell, Perch, Pole, Line, Gaging rod, (or such like instrument) to the Length, Plaine, or Solide measured,

[1.]

* to be certified, either of the length, perimetry, or distance lineall: and this is called, _Mecometrie_. Or

[2.]

* to be certified of the content of any plaine Superficies: whether it be in ground Surueyed, Borde, or Glasse measured, or such like thing: which measuring, is named _Embadometrie_.

[3.]

* Or els to vnderstand the Soliditie, and content of any bodily thing: as of Tymber and Stone, or the content of Pits, Pondes, Wells, Vessels, small & great, of all fashions. Where, of Wine, Oyle, Beere, or Ale vessells, &c, the Measuring, commonly, hath a peculier name: and is called _Gaging_. And the generall name of these Solide measures, is _Stereometrie_.

[+2.+]

Or els, this _vulgar Geometrie_, hath consideration to teach the practiser, how to measure things, with good distance betwene him and the thing measured: and to vnderstand thereby, either

[1.]

* how Farre, a thing seene (on land or water) is from the measurer: and this may be called _Apomecometrie_:

[2.]

Or, how High or depe, aboue or vnder the leuel of the measurers stãding, any thing is, which is sene on land or water, called _Hypsometrie_.

[3.]

* Or, it informeth the measurer, how Broad any thing is, which is in the measurers vew: so it be on Land or Water, situated: and may be called _Platometrie_. Though I vse here to condition, the thing measured, to be on Land, or Water Situated:

[Note.]

yet, know for certaine, that the sundry heigthe of Cloudes, blasing Starres, and of the Mone, may (by these meanes) haue their distances from the earth: and, of the blasing Starres and Mone, the Soliditie (aswell as distances) to be measured: But because, neither these things are vulgarly taught: nor of a common practiser so ready to be executed: I, rather, let such measures be reckened incident to some of our other Artes, dealing with thinges on high, more purposely, then this vulgar Land measuring Geometrie doth: as in _Perspectiue_ and _Astronomie, &c._

Of these Feates (farther applied) is Sprong the Feate of _Geodesie_, or Land Measuring: more cunningly to measure & Suruey Land, Woods, and Waters, a farre of. More cunningly, I say: But God knoweth (hitherto) in these Realmes of England and Ireland (whether through ignorance or fraude, I can not tell, in euery particular)

[Note.]

how great wrong and iniurie hath (in my time) bene committed by vntrue measuring and surueying of Land or Woods, any way. And, this I am sure: that the Value of the difference, betwene the truth and such Surueyes, would haue bene hable to haue foũd (for euer) in eche of our two Vniuersities, an excellent Mathematicall Reader: to eche, allowing (yearly) a hundred Markes of lawfull money of this realme: which, in dede, would seme requisit, here, to be had (though by other wayes prouided for) as well, as, the famous Vniuersitie of Paris, hath two Mathematicall Readers: and eche, two hundreth French Crownes yearly, of the French Kinges magnificent liberalitie onely. Now, againe, to our purpose returning: Moreouer, of the former knowledge Geometricall, are growen the Skills of _Geographie_, _Chorographie_, _Hydrographie_, and _Stratarithmetrie_.

“+‡Geographie‡+ teacheth wayes, by which, in sũdry formes, (as _Sphærike_, _Plaine_ or other), the Situation of Cities, Townes, Villages, Fortes, Castells, Mountaines, Woods, Hauens, Riuers, Crekes, & such other things, vpõ the outface of the earthly Globe (either in the whole, or in some principall mẽber and portion therof cõtayned) may be described and designed, in cõmensurations Analogicall to Nature and veritie: and most aptly to our vew, may be represented.” Of this Arte how great pleasure, and how manifolde commodities do come vnto vs, daily and hourely: of most men, is perceaued. While, some, to beautifie their Halls, Parlers, Chambers, Galeries, Studies, or Libraries with: other some, for thinges past, as battels fought, earthquakes, heauenly fyringes, & such occurentes, in histories mentioned: therby liuely, as it were, to vewe the place, the region adioyning, the distance from vs: and such other circumstances. Some other, presently to vewe the large dominion of the Turke: the wide Empire of the Moschouite: and the litle morsell of ground, where Christendome (by profession) is certainly knowen. Litle, I say, in respecte of the rest. &c. Some, either for their owne iorneyes directing into farre landes: or to vnderstand of other mens trauailes. To conclude, some, for one purpose: and some, for an other, liketh, loueth, getteth, and vseth, Mappes, Chartes, & Geographicall Globes. Of whose vse, to speake sufficiently, would require a booke peculier.

+‡Chorographie‡+ seemeth to be an vnderling, and a twig, of _Geographie_: and yet neuerthelesse, is in practise manifolde, and in vse very ample. “This teacheth Analogically to describe a small portion or circuite of ground, with the contentes: not regarding what commensuration it hath to the whole, or any parcell, without it, contained. But in the territory or parcell of ground which it taketh in hand to make description of, it leaueth out (or vndescribed) no notable, or odde thing, aboue the ground visible. Yea and sometimes, of thinges vnder ground, geueth some peculier marke: or warning: as of Mettall mines, Cole pittes, Stone quarries. &c.” Thus, a Dukedome, a Shiere, a Lordship, or lesse, may be described distinctly. But marueilous pleasant, and profitable it is, in the exhibiting to our eye, and commensuration, the plat of a Citie, Towne, Forte, or Pallace, in true Symmetry: not approching to any of them: and out of Gunne shot. &c. Hereby, the _Architect_ may furnishe him selfe, with store of what patterns he liketh: to his great instruction: euen in those thinges which outwardly are proportioned: either simply in them selues: or respectiuely, to Hilles, Riuers, Hauens, and Woods adioyning. Some also, terme this particular description of places, _Topographie_.

“+‡Hydrographie‡+, deliuereth to our knowledge, on Globe or in Plaine, the perfect Analogicall description of the Ocean Sea coastes, through the whole world: or in the chiefe and principall partes thereof:” with the Iles and chiefe particular places of daungers, conteyned within the boundes, and Sea coastes described: as, of Quicksandes, Bankes, Pittes, Rockes, Races, Countertides, Whorlepooles. &c. This, dealeth with the Element of the water chiefly: as _Geographie_ did principally take the Element of the Earthes description (with his appertenances) to taske. And besides thys, _Hydrographie_, requireth a particular Register of certaine Landmarkes (where markes may be had) from the sea, well hable to be skried, in what point of the Seacumpase they appeare, and what apparent forme, Situation, and bignes they haue, in respecte of any daungerous place in the sea, or nere vnto it, assigned: And in all Coastes, what Mone, maketh full Sea: and what way, the Tides and Ebbes, come and go, the _Hydrographer_ ought to recorde. The Soundinges likewise: and the Chanels wayes: their number, and depthes ordinarily, at ebbe and flud, ought the _Hydrographer_, by obseruation and diligence of _Measuring_, to haue certainly knowen. And many other pointes, are belonging to perfecte _Hydrographie_, and for to make a _Rutter_, by: of which, I nede not here speake: as of the describing, in any place, vpon Globe or Plaine, the 32. pointes of the Compase, truely: (wherof, scarsly foure, in England, haue right knowledge: bycause, the lines therof, are no straight lines, nor Circles.) Of making due proiection of a Sphere in plaine. Of the Variacion of the Compas, from true Northe: And such like matters (of great importance, all) I leaue to speake of, in this place: bycause, I may seame (al ready) to haue enlarged the boundes, and duety of an Hydographer, much more, then any man (to this day) hath noted, or prescribed. Yet am I well hable to proue, all these thinges, to appertaine, and also to be proper to the Hydrographer. The chief vse and ende of this Art, is the Art of Nauigation: but it hath other diuerse vses: euen by them to be enioyed, that neuer lacke sight of land.

+‡Stratarithmetrie‡+, is the Skill, (appertainyng to the warre,) by which a man can set in figure, analogicall to any _Geometricall_ figure appointed, any certaine number or summe of men: of such a figure capable: (by reason of the vsuall spaces betwene Souldiers allowed: and for that, of men, can be made no Fractions. Yet, neuertheles, he can order the giuen summe of men, for the greatest such figure, that of them, cã be ordred) and certifie, of the ouerplus: (if any be) and of the next certaine summe, which, with the ouerplus, will admit a figure exactly proportionall to the figure assigned. By which Skill, also, of any army or company of men: (the figure & sides of whose orderly standing, or array, is knowen) he is able to expresse the iust number of men, within that figure conteined: or (orderly) able to be conteined.

[* Note.]

* And this figure, and sides therof, he is hable to know: either beyng by, and at hand: or a farre of. Thus farre, stretcheth the description and property of _Stratarithmetrie_: sufficient for this tyme and place.

[The difference betwene Stratarithmetrie and Tacticie.]

“It differreth from the Feate _Tacticall_, _De aciebus instruendis._ bycause, there, is necessary the wisedome and foresight, to what purpose he so ordreth the men: and Skillfull hability, also, for any occasion, or purpose, to deuise and vse the aptest and most necessary order, array and figure of his Company and Summe of men.” By figure, I meane: as, either of a _Perfect Square_, _Triangle_, _Circle_, _Ouale_, _long square_, (of the Grekes it is called _Eteromekes_) _Rhombe_, _Rhomboïd_, _Lunular_, _Ryng_, _Serpentine_, and such other Geometricall figures: Which, in warres, haue ben, and are to be vsed: for commodiousnes, necessity, and auauntage &c. And no small skill ought he to haue, that should make true report, or nere the truth, of the numbers and Summes, of footemen or horsemen, in the Enemyes ordring. A farre of, to make an estimate, betwene nere termes of More and Lesse, is not a thyng very rife, among those that gladly would do it.

[I. D. Frende, you will finde it hard, to performe my description of this Feate. But by Chorographie, you may helpe your selfe some what: where the Figures knowne (in Sides and Angles) are not Regular: And where, Resolution into Triangles can serue. &c. And yet you will finde it strange to deale thus generally with Arithmeticall figures: and, that for Battayle ray. Their contentes, differ so much from like Geometricall Figures.]

Great pollicy may be vsed of the Capitaines, (at tymes fete, and in places conuenient) as to vse Figures, which make greatest shew, of so many as he hath: and vsing the aduauntage of the three kindes of vsuall spaces: (betwene footemen or horsemen) to take the largest: or when he would seme to haue few, (beyng many:) contrarywise, in Figure, and space. The Herald, Purseuant, Sergeant Royall, Capitaine, or who soeuer is carefull to come nere the truth herein, besides the Iudgement of his expert eye, his skill of Ordering _Tacticall_, the helpe of his Geometricall instrument: Ring, or Staffe Astronomicall: (commodiously framed for cariage and vse) He may wonderfully helpe him selfe, by perspectiue Glasses. In which, (I trust) our posterity will proue more skillfull and expert, and to greater purposes, then in these dayes, can (almost) be credited to be possible.

Thus haue I lightly passed ouer the Artificiall Feates, chiefly dependyng vpon vulgar _Geometrie_: & commonly and generally reckened vnder the name of _Geometrie_. But there are other (very many) _Methodicall Artes_, which, declyning from the purity, simplicitie, and Immateriality, of our Principall Science of _Magnitudes_: do yet neuertheles vse the great ayde, direction, and Method of the sayd principall Science, and haue propre names, and distinct: both from the Science of _Geometrie_, (from which they are deriued) and one from the other. As +Perspectiue, Astronomie, Musike, Cosmographie, Astrologie, Statike, Anthropographie, Trochilike, Helicosophie, Pneumatithmie, Menadrie, Hypogeiodie, Hydragogie, Horometrie, Zographie, Architecture, Nauigation, Thaumaturgike+ and +Archemastrie+. I thinke it necessary, orderly, of these to giue some peculier descriptions: and withall, to touch some of their commodious vses, and so to make this Preface, to be a little swete, pleasant Nosegaye for you: to comfort your Spirites, beyng almost out of courage, and in despayre, (through brutish brute) Weenyng that _Geometrie_, had but serued for buildyng of an house, or a curious bridge, or the

roufe of Westminster hall, or some witty pretty deuise, or engyn, appropriate to a Carpenter, or a Ioyner &c. That the thing is farre otherwise, then the world, (commonly) to this day, hath demed, by worde and worke, good profe wilbe made.

Among these Artes, by good reason, +‡Perspectiue‡+ ought to be had, ere of _Astronomicall Apparences_, perfect knowledge can be atteyned. And bycause of the prerogatiue of _Light_, beyng the first of _Gods Creatures_: and the eye, the light of our body, and his Sense most mighty, and his organ most Artificiall and _Geometricall_: At _Perspectiue_, we will begyn therfore. +Perspectiue, is an Art Mathematicall, which demonstrateth the maner, and properties, of all Radiations Direct, Broken, and Reflected.+ This Description, or Notation, is brief: but it reacheth so farre, as the world is wyde. It concerneth all Creatures, all Actions, and passions, by Emanation of beames perfourmed. Beames, or naturall lines, (here) I meane, not of light onely, or of colour (though they, to eye, giue shew, witnes, and profe, wherby to ground the Arte vpon) but also of other _Formes_, both _Substantiall_, and _Accidentall_, the certaine and determined actiue Radiall emanations. By this Art (omitting to speake of the highest pointes) we may vse our eyes, and the light, with greater pleasure: and perfecter Iudgement: both of things, in light seen, & of other: which by like order of Lightes Radiations, worke and produce their effectes. We may be ashamed to be ignorant of the cause, why so sundry wayes our eye is deceiued, and abused: as, while the eye weeneth a roũd Globe or Sphere (beyng farre of) to be a flat and plaine Circle, and so likewise iudgeth a plaine Square, to be roũd: supposeth walles parallels, to approche, a farre of: rofe and floure parallels, the one to bend downward, the other to rise vpward, at a little distance from you. Againe, of thinges being in like swiftnes of mouing, to thinke the nerer, to moue faster: and the farder, much slower. Nay, of two thinges, wherof the one (incomparably) doth moue swifter then the other, to deme the slower to moue very swift, & the other to stand: what an error is this, of our eye? Of the Raynbow, both of his Colours, of the order of the colours, of the bignes of it, the place and heith of it, (&c) to know the causes demonstratiue, is it not pleasant, is it not necessary? of two or three Sonnes appearing: of Blasing Sterres: and such like thinges: by naturall causes, brought to passe, (and yet neuertheles, of farder matter, Significatiue) is it not commodious for man to know the very true cause, & occasion Naturall? Yea, rather, is it not, greatly, against the Souerainty of Mans nature, to be so ouershot and abused, with thinges (at hand) before his eyes? as with a Pecockes tayle, and a Doues necke: or a whole ore, in water, holden, to seme broken. Thynges, farre of, to seeme nere: and nere, to seme farre of. Small thinges, to seme great: and great, to seme small. One man, to seme an Army. Or a man to be curstly affrayed of his owne shaddow. Yea, so much, to feare, that, if you, being (alone) nere a certaine glasse, and proffer, with dagger or sword, to foyne at the glasse, you shall suddenly be moued to giue backe (in maner) by reason of an Image,

[A marueilous Glasse.]

appearing in the ayre, betwene you & the glasse, with like hand, sword or dagger, & with like quicknes, foyning at your very eye, likewise as you do at the Glasse. Straunge, this is, to heare of: but more meruailous to behold, then these my wordes

can signifie. And neuerthelesse by demonstration Opticall, the order and cause therof, is certified: euen so, as the effect is consequent. Yea, thus much more, dare I take vpon me, toward the satisfying of the noble courrage, that longeth ardently for the wisedome of Causes Naturall: as to let him vnderstand, that, in London, he may with his owne eyes, haue profe of that, which I haue sayd herein. A Gentleman, (which, for his good seruice, done to his Countrey, is famous and honorable:

[S. W. P.]

and for skill in the Mathematicall Sciences, and Languages, is the Od man of this land. &c.) euen he, is hable: and (I am sure) will, very willingly, let the Glasse, and profe be sene: and so I (here) request him: for the encrease of wisedome, in the honorable: and for the stopping of the mouthes malicious: and repressing the arrogancy of the ignorant. Ye may easily gesse, what I meane. This Art of _Perspectiue_, is of that excellency, and may be led, to the certifying, and executing of such thinges, as no man would easily beleue: without Actuall profe perceiued. I speake nothing of _Naturall Philosophie_, which, without _Perspectiue_, can not be fully vnderstanded, nor perfectly atteined vnto. Nor, of _Astronomie_: which, without _Perspectiue_, can not well be grounded: Nor _Astrologie_, naturally Verified, and auouched. That part hereof, which dealeth with Glasses (which name, Glasse, is a generall name, in this Arte, for any thing, from which, a Beame reboundeth) is called _Catoptrike_: and hath so many vses, both merueilous, and proffitable: that, both, it would hold me to long, to note therin the principall conclusions, all ready knowne: And also (perchaunce) some thinges, might lacke due credite with you: And I, therby, to leese my labor: and you, to slip into light Iudgement,

[* □]

Before you haue learned sufficiently the powre of Nature and Arte.

Now, to procede: +‡Astronomie‡, is an Arte Mathematicall, which demonstrateth the distance, magnitudes, and all naturall motions, apparences, and passions propre to the Planets and fixed Sterres: for any time past, present and to come: in respect of a certaine Horizon, or without respect of any Horizon.+ By this Arte we are certified of the distance of the Starry Skye, and of eche _Planete_ from the Centre of the Earth: and of the greatnes of any Fixed starre sene, or _Planete_, in respect of the Earthes greatnes. As, we are sure (by this Arte) that the Solidity, Massines and Body of the _Sonne_, conteineth the quantitie of the whole Earth and Sea, a hundred thre score and two times, lesse by ⅛ one eight parte of the earth. But the Body of the whole earthly globe and Sea, is bigger then the body of the Mone, three and forty times lesse by ⅛ of the Mone. Wherfore the _Sonne_ is bigger then the _Mone_, 7000 times, lesse, by 59 39/64 that is, precisely 6940 25/64 bigger then the _Mone_. And yet the vnskillfull man, would iudge them a like bigge. Wherfore, of Necessity, the one is much farder from vs, then the other. The _Sonne_, when he is fardest from the earth (which, now, in our age, is, when he is in the 8. degree, of Cancer) is, 1179 Semidiameters of the Earth, distante. And the _Mone_ when she is fardest from the earth, is 68 Semidiameters of the earth and ⅓ The nerest, that the _Mone_ commeth

to the earth, is Semidiameters 52¼ The distance of the Starry Skye is, frõ vs, in Semidiameters of the earth 20081½ Twenty thousand fourescore, one, and almost a halfe. Subtract from this, the _Mones_ nerest distance, from the Earth: and therof remaineth Semidiameters of the earth 20029¼ Twenty thousand nine and twenty and a quarter.

[Note.]

So thicke is the heauenly Palace, that the _Planetes_ haue all their exercise in, and most meruailously perfourme the Commaũdement and Charge to them giuen by the omnipotent Maiestie of the king of kings. This is that, which in _Genesis_ is called _Ha Rakia_. Consider it well. The Semidiameter of the earth, cõteineth of our common miles 3436 4/11 three thousand, foure hundred thirty six and foure eleuenth partes of one myle: Such as the whole earth and Sea, round about, is 21600. One and twenty thousand six hundred of our myles. Allowyng for euery degree of the greatest circle, thre score myles. Now if you way well with your selfe but this litle parcell of frute _Astronomicall_, as concerning the bignesse, Distances of _Sonne_, _Mone_, _Sterry Sky_, and the huge massines of _Ha Rakia_, will you not finde your Consciences moued, with the kingly Prophet, to sing the confession of Gods Glory, and say, +_The Heauens declare the glory of God, and the Firmament =[Ha Rakia]= sheweth forth the workes of his handes_+. And so forth, for those fiue first staues, of that kingly Psalme. Well, well, It is time for some to lay hold on wisedome, and to Iudge truly of thinges: and notso to expound the Holy word, all by Allegories: as to Neglect the wisedome, powre and Goodnes of God, in, and by his Creatures, and Creation to be seen and learned. By parables and Analogies of whose natures and properties, the course of the Holy Scripture, also, declareth to vs very many Mysteries. The whole Frame of Gods Creatures, (which is the whole world,) is to vs, a bright glasse: from which, by reflexion, reboundeth to our knowledge and perceiuerance, Beames, and Radiations: representing the Image of his Infinite goodnes, Omnipotẽcy, and wisedome. And we therby, are taught and persuaded to Glorifie our Creator, as God: and be thankefull therfore. Could the Heathenistes finde these vses, of these most pure, beawtifull, and Mighty Corporall Creatures: and shall we, after that the true _Sonne_ of rightwisenesse is risen aboue the _Horizon_, of our temporall _Hemisphærie_, and hath so abundantly streamed into our hartes, the direct beames of his goodnes, mercy, and grace: Whose heat All Creatures feele: Spirituall and Corporall: Visible and Inuisible. Shall we (I say) looke vpon the _Heauen_, _Sterres_, and _Planets_, as an Oxe and an Asse doth: no furder carefull or inquisitiue, what they are: why were they Created, How do they execute that they were Created for? Seing, All Creatures, were for our sake created: and both we, and they, Created, chiefly to glorifie the Almighty Creator: and that, by all meanes, to vs possible. _Nolite ignorare_ (saith _Plato in Epinomis_) _Astronomiam, Sapientissimũ quiddam esse._ +_Be ye not ignorant, Astronomie to be a thyng of excellent wisedome._+ _Astronomie_, was to vs, from the beginning commended, and in maner commaunded by God him selfe. In asmuch as he made the _Sonne_, _Mone_, and _Sterres_, to be to vs, for _Signes_, and knowledge of Seasons, and for Distinctions of Dayes, and yeares. Many wordes nede not. But I wish, euery man should way this word, _Signes_. And besides that, conferre it also with the tenth Chapter of _Hieremie_. And though Some

thinke, that there, they haue found a rod: Yet Modest Reason, will be indifferent Iudge, who ought to be beaten therwith, in respect of our purpose. Leauing that: I pray you vnderstand this: that without great diligence of Obseruation, examination and Calculation, their periods and courses (wherby _Distinction_ of Seasons, yeares, and New Mones might precisely be knowne) could not exactely be certified. Which thing to performe, is that _Art_, which we here haue Defined to be _Astronomie_. Wherby, we may haue the distinct Course of Times, dayes, yeares, and Ages: aswell for Consideratiõ of Sacred Prophesies, accomplished in due time, foretold: as for high Mysticall Solemnities holding: And for all other humaine affaires, Conditions, and couenantes, vpon certaine time, betwene man and man: with many other great vses: Wherin, (verely), would be great incertainty, Confusion, vntruth, and brutish Barbarousnes: without the wonderfull diligence and skill of this Arte: continually learning, and determining Times, and periodes of Time, by the Record of the heauenly booke, wherin all times are written: and to be read with an _Astronomicall staffe_, in stede of a festue.

+‡Musike‡+, of Motion, hath his Originall cause: Therfore, after the motions most swift, and most Slow, which are in the Firmament, of Nature performed: and vnder the _Astronomers Consideration_: now I will Speake of an other kinde of _Motion_, producing sound, audible, and of Man numerable. _Musike_ I call here that _Science_, which of the Grekes is called _Harmonice_. Not medling with the Controuersie betwene the auncient _Harmonistes_, and _Canonistes_. +Musike is a Mathematicall Science, which teacheth, by sense and reason, perfectly to iudge, and order the diuersities of soundes, hye and low.+ _Astronomie_ and _Musike_ are Sisters, saith _Plato_. As, for _Astronomie_, the eyes: So, for _Harmonious Motion_, the eares were made. But as _Astronomie_ hath a more diuine Contemplation, and cõmodity, then mortall eye can perceiue: So, is _Musike_ to be considered,

[1.]

that the * Minde may be preferred, before the eare. And from audible sound, we ought to ascende, to the examination: which numbers are _Harmonious_, and which not. And why, either, the one are: or the other are not. I could at large,

[2.]

in the heauenly * motions and distances, describe a meruailous Harmonie, of _Pythagoras_ Harpe

[3.]

with eight stringes. Also, somwhat might be sayd of _Mercurius_ * two Harpes,

[4.]

eche of foure Stringes Elementall. And very straunge matter, might be alledged of the _Harmonie_,

[5.]

to our * Spirituall part appropriate. As in _Ptolomaus_ third boke, in the fourth and sixth Chapters may appeare. *

[6.]

And what is the cause of the apt bonde, and frendly felowship, of the Intellectuall and Mentall part of vs, with our grosse & corruptible body: but a certaine Meane, and _Harmonious Spiritualitie, with both participatyng, & of both (in a maner) resultynge In

[7.]

the * Tune of Mans voyce, and also

[8.]

* the sound of Instrument_, what might be sayd, of _Harmonie_: No common Musicien would lightly beleue.

[I. D. Read in Aristotle his 8. booke of Politikes: the 5, 6, and 7. chapters. Where you shall haue some occasion farder to thinke of Musike, than commonly is thought.]

But of the sundry Mixture (as I may terme it) and concurse, diuerse collation, and Application of these _Harmonies_: as of thre, foure, fiue, or mo: Maruailous haue the effectes ben: and yet may be founde, and produced the like: with some proportionall consideration for our time, and being: in respect of the State, of the thinges then: in which, and by which, the wondrous effectes were wrought. _Democritus_ and _Theophrastus_ affirmed, that, by _Musike_, griefes and diseases of the Minde, and body might be cured, or inferred. And we finde in Recorde, that _Terpander_, _Arion_, _Ismenias_, _Orpheus_, _Amphion_, _Dauid_, _Pythagoras_, _Empedocles_, _Asclepiades_ and _Timotheus_, by _Harmonicall_ Consonãcy, haue done, and brought to pas, thinges, more then meruailous, to here of. Of them then, making no farder discourse, in this place: Sure I am, that Common _Musike_, commonly vsed, is found to the _Musiciens_ and Hearers, to be so Commodious and pleasant, That if I would say and dispute, but thus much: That it were to be otherwise vsed, then it is, I should finde more repreeuers, then I could finde priuy, or skilfull of my meaning. In thinges therfore euident, and better knowen, then I can expresse: and so allowed and liked of, (as I would wish, some other thinges, had the like hap) I will spare to enlarge my lines any farder, but consequently follow my purpose.

+‡Of Cosmographie‡+, I appointed briefly in this place, to geue you some intelligence. +Cosmographie, is the whole and perfect description of the heauenly, and also elementall parte of the world, and their homologall application, and mutuall collation necessarie.+ This Art, requireth _Astronomie_, _Geographie_, _Hydrographie_ and _Musike_. Therfore, it is no small Arte, nor so simple, as in common practise, it is (slightly) considered. This matcheth Heauen, and the Earth, in one frame, and aptly applieth parts Correspõdent: So, as, the Heauenly Globe, may (in practise) be duely described vpon the Geographicall, and Hydrographicall Globe. And there, for vs to consider an _Æquonoctiall Circle_, _an Ecliptike line_, _Colures_, _Poles_, _Sterres_ in their true Longitudes, Latitudes, Declinations, and Verticalitie: also Climes, and Parallels: and by an _Horizon_ annexed, and reuolution of the earthly Globe (as the Heauen, is, by the _Primouant_, caried about in 24. æquall Houres) to learne the Risinges and Settinges of Sterres (of _Virgill_ in his _Georgikes_: of

Hesiod: of _Hippocrates_ in his _Medicinall Sphære_, to Perdicca King of the Macedonians: of _Diocles_, to King _Antigonus_, and of other famous _Philosophers_ prescribed) a thing necessary, for due manuring of the earth, for _Nauigation_, for the Alteration of mans body: being, whole, Sicke, wounded, or brused. By the Reuolution, also, or mouing of the Globe Cosmographicall, the Rising and Setting of the Sonne: the Lengthes, of dayes and nightes: the Houres and times (both night and day) are knowne: with very many other pleasant and necessary vses: Wherof, some are knowne: but better remaine, for such to know and vse:

[□]

who of a sparke of true fire, can make a wonderfull bonfire, by applying of due matter, duely.

+‡Of Astrologie‡+, here I make an Arte, seuerall from _Astronomie_: not by new deuise, but by good reason and authoritie: for, +Astrologie, is an Arte Mathematicall, which reasonably demonstrateth the operations and effectes, of the naturall beames, of light, and secrete influence: of the Sterres and Planets: in euery element and elementall body: at all times, in any Horizon assigned.+ This Arte is furnished with many other great Artes and experiences: As with perfecte _Perspectiue_, _Astronomie_, _Cosmographie_, _Naturall Philosophie_ of the 4. Elementes, the Arte of Graduation, and some good vnderstãding in _Musike_: and yet moreouer, with an other great Arte, hereafter following, though I, here, set this before, for some considerations me mouing. Sufficient (you see) is the stuffe, to make this rare and secrete Arte, of: and hard enough to frame to the Conclusion Syllogisticall. Yet both the manifolde and continuall trauailes of the most auncient and wise Philosophers, for the atteyning of this Arte: and by examples of effectes, to confirme the same: hath left vnto vs sufficient proufe and witnesse: and we, also, daily may perceaue, That mans body, and all other Elementall bodies, are altered, disposed, ordred, pleasured, and displeasured, by the Influentiall working of the _Sunne_, _Mone_, and the other Starres and Planets. And therfore, sayth _Aristotle_, in the first of his _Meteorologicall_ bookes, in the second Chapter: _Est autem necessariò Mundus iste, supernis lationibus ferè continuus. Vt, inde, vis eius vniuersa regatur. Ea siquidem Causà prima putanda omnibus est, vnde motus principium existit._ That is: +_This =[Elementall]= World is of necessitie, almost, next adioyning, to the heauenly motions: That, from thence, all his vertue or force may be gouerned. For, that is to be thought the first Cause vnto all: from which, the beginning of motion, is._+ And againe, in the tenth Chapter. _Oportet igitur & horum principia sumamus, & causas omnium similiter. Principium igitur vt mouens, præcipuum[que] & omnium primum, Circulus ille est, in quo manifeste Solis latio, &c._ And so forth. His _Meteorologicall_ bookes, are full of argumentes, and effectuall demonstrations, of the vertue, operation, and power of the heauenly bodies, in and vpon the fower Elementes, and other bodies, of them (either perfectly, or vnperfectly) composed. And in his second booke, _De Generatione & Corruptione_, in the tenth Chapter. _Quocirca & prima latio, Ortus & Interitus causa non est: Sed obliqui Circuli latio: ea nam[que] & continua est, & duobus motibus fit:_ In Englishe, thus. +_Wherefore the vppermost motion, is not the cause of Generation and Corruption, but the motion of the Zodiake:

for, that, both, is continuall, and is caused of two mouinges._+ And in his second booke, and second Chapter of hys _Physikes_. _Homo nam[que] generat hominem, at[que] Sol._ +_For Man (sayth he) and the Sonne, are cause of mans generation._+ Authorities may be brought, very many: both of 1000. 2000. yea and 3000. yeares Antiquitie: of great _Philosophers_, _Expert_, _Wise_, and godly men, for that Conclusion: which, daily and hourely, we men, may discerne and perceaue by sense and reason: All beastes do feele, and simply shew, by their actions and passions, outward and inward: All Plants, Herbes, Trees, Flowers, and Fruites. And finally, the Elementes, and all thinges of the Elementes composed, do geue Testimonie (as _Aristotle_ sayd) that theyr +_Whole Dispositions, vertues, and naturall motions, depend of the Actiuitie of the heauenly motions and Influences. Whereby, beside the specificall order and forme, due to euery seede: and beside the Nature, propre to the Indiuiduall Matrix, of the thing produced: What shall be the heauenly Impression, the perfect and circumspecte Astrologien hath to Conclude._+ Not onely (by _Apotelesmes_) τὸ ὁτὶ]. but by Naturall and Mathematicall demonstration τὸ διότι. Whereunto, what Sciences are requisite (without exception) I partly haue here warned: And in my _Propædeumes_ (besides other matter there disclosed) I haue Mathematically furnished vp the whole Method: To this our age, not so carefully handled by any, that euer I saw, or heard of. I was,

[* Anno. 1548 and 1549. in Louayn.]

(for * 21. yeares ago) by certaine earnest disputations, of the Learned _Gerardus Mercator_, and _Antonius Gogaua_, (and other,) therto so prouoked: and (by my constant and inuincible zeale to the veritie) in obseruations of Heauenly Influencies (to the Minute of time,) than, so diligent: And chiefly by the Supernaturall influence, from the Starre of Iacob, so directed: That any Modest and Sober Student, carefully and diligently seking for the Truth, will both finde & cõfesse, therin, to be the Veritie, of these my wordes: And also become a Reasonable Reformer, of three Sortes of people: about these Influentiall Operations, greatly erring from the truth.

[Note.]

Wherof, the one, is +Light Beleuers+, the other, +Light Despisers+, and the third +Light Practisers+. The first, & most cõmon Sort, thinke the Heauen and Sterres, to be answerable to any their doutes or desires:

[1.]

which is not so: and, in dede, they, to much, ouer reache. The Second sorte thinke no Influentiall vertue (frõ the heauenly bodies) to beare any Sway in Generation

[2.]

and Corruption, in this Elementall world. And to the _Sunne_, _Mone_ and _Sterres_ (being so many, so pure, so bright, so wonderfull bigge, so farre in distance, so manifold in their motions, so constant in their periodes. &c.) they assigne a sleight, simple office or two, and so allow vnto thẽ (according to their capacities) as much vertue, and power Influentiall, as to the Signe of the _Sunne_, _Mone_, and seuen Sterres, hanged vp (for Signes) in London, for distinction of houses, & such grosse

helpes, in our worldly affaires: And they vnderstand not (or will not vnderstand) of the other workinges, and vertues of the Heauenly _Sunne_, _Mone_, and _Sterres_: not so much, as the Mariner, or Husband man: no, not so much, as the _Elephant_ doth, as the _Cynocephalus_, as the Porpentine doth: nor will allow these perfect, and incorruptible mighty bodies, so much vertuall Radiation, & Force, as they see in a litle peece of a _Magnes stone_: which, at great distance, sheweth his operation. And perchaunce they thinke, the Sea & Riuers (as the Thames) to be some quicke thing, and so to ebbe, and flow, run in and out, of them selues, at their owne fantasies. God helpe, God helpe. Surely, these men, come to short: and either are to dull: or willfully blind: or, perhaps, to malicious. The third man, is the common and vulgare _Astrologien_, or Practiser: who, being not duely, artificially, and perfectly

[3.]

furnished: yet, either for vaine glory, or gayne: or like a simple dolt, & blinde Bayard, both in matter and maner, erreth: to the discredit of the _Wary_, and modest _Astrologien_: and to the robbing of those most noble corporall Creatures, of their Naturall Vertue: being most mighty: most beneficiall to all elementall Generation, Corruption and the appartenances: and most Harmonious in their Monarchie: For which thinges, being knowen, and modestly vsed: we might highly, and continually glorifie God, with the princely Prophet, saying. +_The Heauens declare the Glorie of God: who made the Heauẽs in his wisedome: who made the Sonne, for to haue dominion of the day: the Mone and Sterres to haue dominion of the nyght: whereby, Day to day vttereth talke: and night, to night declareth knowledge. Prayse him, all ye Sterres, and Light. Amen._+

In order, now foloweth, of +‡Statike‡+, somewhat to say, what we meane by that name: and what commodity, doth, on such Art, depend. +Statike, is an Arte Mathematicall, which demonstrateth the causes of heauynes, and lightnes of all thynges: and of motions and properties, to heauynes and lightnes, belonging.+ And for asmuch as, by the Bilanx, or Balance (as the chief sensible Instrument,) Experience of these demonstrations may be had: we call this Art, _Statike:_ that is, _the Experimentes of the Balance_. Oh, that men wist, what proffit, (all maner of wayes) by this Arte might grow, to the hable examiner, and diligent practiser. “Thou onely, knowest all thinges precisely (O God) who hast made weight and Balance, thy Iudgement: who hast created all thinges in _Number, Waight, and Measure_: and hast wayed the mountaines and hils in a Balance: who hast peysed in thy hand, both Heauen and earth. We therfore warned by the Sacred word, to Consider thy Creatures: and by that consideration, to wynne a glyms (as it were,) or shaddow of perceiuerance, that thy wisedome, might, and goodnes is infinite, and vnspeakable, in thy Creatures declared: And being farder aduertised, by thy mercifull goodnes, that, three principall wayes, were, of the, vsed in Creation of all thy Creatures, namely, _Number_, _Waight_ and _Measure_, And for as much as, of _Number_ and _Measure_, the two Artes (auncient, famous, and to humaine vses most necessary,) are, all ready, sufficiently knowen and extant: This third key, we beseche thee (through thy accustomed goodnes,) that it may come to the nedefull and sufficient knowledge, of such thy Seruauntes, as in thy workemanship, would gladly finde, thy

true occasions (purposely of the vsed) whereby we should glorifie thy name, and shew forth (to the weaklinges in faith) thy wondrous wisedome and Goodnes. Amen.”

Meruaile nothing at this pang (godly frend, you Gentle and zelous Student.) An other day, perchaunce, you will perceiue, what occasion moued me. Here, as now, I will giue you some ground, and withall some shew, of certaine commodities, by this Arte arising. And bycause this Arte is rare, my wordes and practises might be to darke: vnleast you had some light, holden before the matter: and that, best will be, in giuing you, out of _Archimedes_ demonstrations, a few principal Conclusions, as foloweth.

+1.+

+The Superficies of euery Liquor, by it selfe consistyng, and in quyet, is Sphæricall: the centre whereof, is the same, which is the centre of the Earth.+

+2.+

+If Solide Magnitudes, being of the same bignes, or quãtitie, that any Liquor is, and hauyng also the same Waight: be let downe into the same Liquor, they will settle downeward, so, that no parte of them, shall be aboue the Superficies of the Liquor: and yet neuertheles, they will not sinke vtterly downe, or drowne.+

+3.+

+If any Solide Magnitude beyng Lighter then a Liquor, be let downe into the same Liquor, it will settle downe, so farre into the same Liquor, that so great a quantitie of that Liquor, as is the parte of the Solid Magnitude, settled downe into the same Liquor: is in Waight, æquall, to the waight of the whole Solid Magnitude.+

+4.+

+Any Solide Magnitude, Lighter then a Liquor, forced downe into the same Liquor, will moue vpward, with so great a power, by how much, the Liquor hauyng æquall quantitie to the whole Magnitude, is heauyer then the same Magnitude.+

+5.+

+Any Solid Magnitude, heauyer then a Liquor, beyng let downe into the same Liquor, will sinke downe vtterly: And wilbe in that Liquor, Lighter by so much, as is the waight or heauynes of the Liquor, hauing bygnes or quantitie, æquall to the Solid Magnitude.+

+6.+

[I. D. The Cutting of a Sphære according to any proportion assigned may by this proposition be done Mechanically by tempering Liquor to a certayne waight in respect of the waight of the Sphære therein Swymming.]

+If any Solide Magnitude, Lighter then a Liquor, be let downe into the same Liquor, the waight of the same Magnitude, will be, to the Waight of the Liquor. (Which is æquall in quantitie to the whole Magnitude,) in that proportion, that the parte, of the Magnitude settled downe, is to the whole Magnitude.+

By these verities, great Errors may be reformed, in Opinion of the Naturall Motion of thinges, Light and Heauy. Which errors, are in Naturall Philosophie (almost) of all mẽ allowed: to much trusting to Authority: and false Suppositions. As, +Of any two bodyes, the heauyer, to moue downward faster then the lighter.+

[A common error, noted.]

This error, is not first by me, Noted: but by one _Iohn Baptist de Benedictis_. The chief of his propositions, is this: which seemeth a Paradox.

+If there be two bodyes of one forme, and of one kynde, æquall in quantitie or vnæquall,

[A paradox.]

they will moue by æquall space, in æquall tyme: So that both theyr mouynges be in ayre, or both in water: or in any one Middle.+

Hereupon, in the feate of +Gunnyng+,

[N. T.]

certaine good discourses (otherwise) may receiue great amendement, and furderance.

[The wonderfull vse of these Propositions.]

In the entended purpose, also, allowing somwhat to the imperfection of Nature: not aunswerable to the precisenes of demonstration. Moreouer, by the foresaid propositions (wisely vsed.) The Ayre, the water, the Earth, the Fire, may be nerely, knowen, how light or heauy they are (Naturally) in their assigned partes: or in the whole. And then, to thinges Elementall, turning your practise: you may deale for the proportion of the Elementes, in the thinges Compounded. Then, to the proportions of the Humours in Man: their waightes: and the waight of his bones, and flesh. &c. Than, by waight, to haue consideration of the Force of man, any maner of way: in whole or in part. Then, may you, of Ships water drawing, diuersly, in the Sea and in fresh water, haue pleasant consideration: and of waying vp of any thing, sonken in Sea or in fresh water &c. And (to lift vp your head a loft:) by waight, you may, as precisely, as by any instrument els, measure the Diameters of _Sonne_ and _Mone. &c._ Frende, I pray you, way these thinges, with the iust Balance of Reason. And you will finde Meruailes vpon Meruailes: And esteme one Drop of Truth (yea in Naturall Philosophie) more worth, then whole Libraries of Opinions, vndemonstrated: or not aunswering to Natures Law, and your experience. Leauing these thinges, thus: I will giue you two or three, light practises, to great purpose: and so finish my Annotation _Staticall_. In Mathematicall matters, by the Mechaniciens ayde, we will behold, here, the Commodity of waight.

[The practise Staticall, to know the proportion, betwene the Cube, and the Sphære.]

Make a Cube, of any one Vniforme: and through like heauy stuffe: of the same Stuffe, make a Sphære or Globe, precisely, of a Diameter æquall to the Radicall side of the Cube. Your stuffe, may be wood, Copper, Tinne, Lead, Siluer. &c. (being, as I sayd, of like nature, condition, and like waight throughout.) And you may, by Say Balance, haue prepared a great number of the smallest waightes: which, by those Balance can be discerned or tryed: and so, haue proceded to make you a perfect Pyle, company & Number of waightes: to the waight of six, eight, or twelue pound waight: most diligently tryed, all. And of euery one, the Content knowen, in your least waight, that is wayable. [They that can not haue these waightes of precisenes: may, by Sand, Vniforme, and well dusted, make them a number of waightes, somewhat nere precisenes: by halfing euer the Sand: they shall, at length, come to a least common waight. Therein, I leaue the farder matter, to their discretion, whom nede shall pinche.] The _Venetians_ consideration of waight, may seme precise enough: by eight descentes progressionall, * halfing, from a grayne.

[I. D. * For, so, haue you .256. partes of a Graine.]

Your Cube, Sphære, apt Balance, and conuenient waightes, being ready: fall to worke.. First, way your Cube. Note the Number of the waight. Way, after that, your Sphære. Note likewise, the Nũber of the waight. If you now find the waight of your Cube, to be to the waight of the Sphære, as 21. is to 11: Then you see, how the Mechanicien and _Experimenter_, without Geometrie and Demonstration, are (as nerely in effect) tought the proportion of the Cube to the Sphere: as I haue demonstrated it, in the end of the twelfth boke of _Euclide_. Often, try with the same Cube and Sphære. Then, chaunge, your Sphære and Cube, to an other matter: or to an other bignes: till you haue made a perfect vniuersall Experience of it. Possible it is, that you shall wynne to nerer termes, in the proportion.

When you haue found this one certaine Drop of Naturall veritie, procede on, to Inferre, and duely to make assay, of matter depending. As, bycause it is well demonstrated, that a Cylinder, whose heith, and Diameter of his base, is æquall to the Diameter of the Sphære, is Sesquialter to the same Sphære (that is, as 3. to 2:) To the number of the waight of the Sphære, adde halfe so much, as it is: and so haue you the number of the waight of that Cylinder. Which is also Comprehended of our former Cube: So, that the base of that Cylinder, is a Circle described in the Square, which is the base of our Cube. But the Cube and the Cylinder, being both of one heith, haue their Bases in the same proportion, in the which, they are, one to an other, in their Massines or Soliditie. But, before, we haue two numbers, expressing their Massines, Solidities, and Quantities, by waight: wherfore,

[* =The proportion of the Square to the Circle inscribed.=]

we haue * the proportion of the Square, to the Circle, inscribed in the same Square. And so are we fallen into the knowledge sensible, and Experimentall of _Archimedes_ great Secret: of him, by great trauaile of minde, sought and found. Wherfore, to any Circle giuen, you can giue a Square æquall:

[* =The Squaring of the Circle, Mechanically.=]

* as I haue taught, in my Annotation, vpon the first proposition of the twelfth boke, And likewise, to any Square giuen, you may giue a Circle æquall:

[* =To any Square geuen, to geue a Circle, equall.=]

* If you describe a Circle, which shall be in that proportion, to your Circle inscribed, as the Square is to the same Circle: This, you may do, by my Annotations, vpon the second proposition of the twelfth boke of _Euclide_, in my third Probleme there. Your diligence may come to a proportion, of the Square to the Circle inscribed, nerer the truth, then is the proportion of 14. to 11. And consider, that you may begyn at the Circle and Square, and so come to conclude of the Sphære, & the Cube, what their proportion is: as now, you came from the Sphære to the Circle. For, of Siluer, or Gold, or Latton Lamyns or plates (thorough one hole drawẽ, as the maner is) if you make a Square figure & way it: and then, describing theron, the Circle inscribed: & cut of, & file away, precisely (to the Circle) the ouerplus of the Square: you shall then, waying your Circle, see, whether the waight of the Square, be to your Circle, as 14. to 11. As I haue Noted, in the beginning of _Euclides_ twelfth boke. &c. after this resort to my last proposition, vpon the last of the twelfth. And there, helpe your selfe, to the end. And, here, Note this, by the way.

[Note Squaring of the Circle without knowledge of the proportion betwene Circumference and Diameter.]

That we may Square the Circle, without hauing knowledge of the proportion, of the Circumference to the Diameter: as you haue here perceiued. And otherwayes also, I can demonstrate it. So that, many haue cumberd them selues superfluously, by trauailing in that point first, which was not of necessitie, first: and also very intricate. And easily, you may, (and that diuersly) come to the knowledge of the Circumference: the Circles Quantitie, being first knowen. Which thing, I leaue to your consideration: making hast to despatch an other Magistrall Probleme: and to bring it, nerer to your knowledge, and readier dealing with, then the world (before this day,) had it for you, that I can tell of. And that is, _A Mechanicall Dubblyng of the Cube: &c._ Which may, thus, be done:

[To Dubble the Cube redily: by Art Mechanicall: depending vppon Demonstration Mathematicall.]

+Make of Copper plates, or Tyn plates, a foursquare vpright Pyramis, or a Cone: perfectly fashioned in the holow, within. Wherin, let great diligence be vsed, to approche (as nere as may be) to the Mathematicall perfection of those figures. At their bases, let them be all open: euery where, els, most close, and iust to. From the vertex, to the Circumference of the base of the Cone: & to the sides of the base of the Pyramis:+

[=I. D.= =The 4. sides of this Pyramis must be 4. Isosceles Triangles alike and æquall.=]

+Let 4. straight lines be drawen, in the inside of the Cone and Pyramis: makyng at their fall, on the perimeters of the bases, equall angles on both sides them selues, with the sayd perimeters. These 4. lines (in the Pyramis: and as many, in the Cone)

diuide: one, in 12. æquall partes: and an other, in 24. an other, in 60, and an other, in 100. (reckenyng vp from the vertex.) Or vse other numbers of diuision, as experience shall teach you.+

[=I. D.= =* In all workinges with this Pyramis or Cone, Let their Situations be in all Pointes and Conditions, alike, or all one: while you are about one Worke. Els you will erre.=]

+Then, * set your Cone or Pyramis, with the vertex downward, perpendicularly, in respect of the Base. (Though it be otherwayes, it hindreth nothyng.) So let thẽ most stedily be stayed.+ Now, if there be a Cube, which you wold haue Dubbled. Make you a prety Cube of Copper, Siluer, Lead, Tynne, Wood, Stone, or Bone. Or els make a hollow Cube, or Cubik coffen, of Copper, Siluer, Tynne, or Wood &c. These, you may so proportiõ in respect of your Pyramis or Cone, that the Pyramis or Cone, will be hable to conteine the waight of them, in water, 3. or 4. times: at the least: what stuff so euer they be made of. Let not your Solid angle, at the vertex, be to sharpe: but that the water may come with ease, to the very vertex, of your hollow Cone or Pyramis. Put one of your Solid Cubes in a Balance apt: take the waight therof exactly in water. Powre that water, (without losse) into the hollow Pyramis or Cone, quietly. Marke in your lines, what numbers the water Cutteth: Take the waight of the same Cube againe: in the same kinde of water, which you had before:

[=I. D.= =* Consider well whan you must put your waters togyther: and whan, you must empty your first water, out of your Pyramis or Cone. Els you will erre.=]

put that* also, into the Pyramis or Cone, where you did put the first. Marke now againe, in what number or place of the lines, the water Cutteth them. Two wayes you may conclude your purpose: it is to wete, either by numbers or lines. By numbers: as, if you diuide the side of your Fundamentall Cube into so many æquall partes, as it is capable of, conueniently, with your ease, and precisenes of the diuision. For, as the number of your first and lesse line (in your hollow Pyramis or Cone,) is to the second or greater (both being counted from the vertex) so shall the number of the side of your Fundamentall Cube, be to the nũber belonging to the Radicall side, of the Cube, dubble to your Fundamentall Cube: Which being multiplied Cubik wise, will sone shew it selfe, whether it be dubble or no, to the Cubik number of your Fundamentall Cube. By lines, thus: As your lesse and first line, (in your hollow Pyramis or Cone,) is to the second or greater, so let the Radical side of your Fundamẽtall Cube, be to a fourth proportionall line, by the 12. proposition, of the sixth boke of _Euclide_. Which fourth line, shall be the Rote Cubik, or Radicall side of the Cube, dubble to your Fundamentall Cube: which is the thing we desired.

[God be thanked for this Inuention, & the fruite ensuing.]

For this, may I (with ioy) say, EYPHKA, EYPHKA, EYPHKA: thanking the holy and glorious Trinity: hauing greater cause therto, then

[* Vitruuius. Lib. 9. Cap. 3.]

* _Archimedes_ had (for finding the fraude vsed in the Kinges Crowne, of Gold): as all men may easily Iudge: by the diuersitie of the frute following of the one, and the

other. Where I spake before, of a hollow Cubik Coffen: the like vse, is of it: and without waight. Thus. Fill it with water, precisely full, and poure that water into your Pyramis or Cone. And here note the lines cutting in your Pyramis or Cone. Againe, fill your coffen, like as you did before. Put that Water, also, to the first. Marke the second cutting of your lines. Now, as you proceded before, so must you here procede.

[* Note.]

* And if the Cube, which you should Double, be neuer so great: you haue, thus, the proportion (in small) betwene your two litle Cubes: And then, the side, of that great Cube (to be doubled) being the third, will haue the fourth, found, to it proportionall: by the 12. of the sixth of Euclide.

[Note, as concerning the Sphæricall Superficies of the Water.]

Note, that all this while, I forget not my first Proposition Staticall, here rehearsed: that, the Superficies of the water, is Sphæricall. Wherein, vse your discretion: to the first line, adding a small heare breadth, more: and to the second, halfe a heare breadth more, to his length. For, you will easily perceaue, that the difference can be no greater, in any Pyramis or Cone, of you to be handled. Which you shall thus trye. _For finding the swelling of the water aboue leuell._

[□]

“Square the Semidiameter, from the Centre of the earth, to your first Waters Superficies. Square then, halfe the Subtendent of that watry Superficies (which Subtendent must haue the equall partes of his measure, all one, with those of the Semidiameter of the earth to your watry Superficies): Subtracte this square, from the first: Of the residue, take the Rote Square. That Rote, Subtracte from your first Semidiameter of the earth to your watry Superficies: that, which remaineth, is the heith of the water, in the middle, aboue the leuell.” Which, you will finde, to be a thing insensible. And though it were greatly sensible, *

[* Note.]

yet, by helpe of my sixt Theoreme vpon the last Proposition of Euclides twelfth booke, noted: you may reduce all, to a true Leuell. But, farther diligence, of you is to be vsed, against accidentall causes of the waters swelling: as by hauing (somwhat) with a moyst Sponge, before, made moyst your hollow Pyramis or Cone, will preuent an accidentall cause of Swelling, &c. Experience will teach you abundantly: with great ease, pleasure, and cõmoditie.

Thus, may you Double the Cube Mechanically, Treble it, and so forth, in any proportion.

[Note this Abridgement of Dubbling the Cube. &c.]

Now will I Abridge your paine, cost, and Care herein. Without all preparing of your Fundamentall Cubes: you may (alike) worke this Conclusion. For, that, was rather a kinde of Experimentall demõstration, then the shortest way: and all, vpon one Mathematicall Demonstration depending. “Take water (as much as conueniently will serue your turne: as I warned before of your Fundamentall Cubes bignes) Way it

precisely. Put that water, into your Pyramis or Cone. Of the same kinde of water, then take againe, the same waight you had before: put that likewise into the Pyramis or Cone. For, in eche time, your marking of the lines, how the Water doth cut them, shall geue you the proportion betwen the Radicall sides, of any two Cubes, wherof the one is Double to the other: working as before I haue taught you:

[* □ Note.]

* sauing that for you Fundamentall Cube his Radicall side: here, you may take a right line, at pleasure.”

Yet farther proceding with our droppe of Naturall truth:

[To giue Cubes one to the other in any proportion, Rationall or Irrationall.]

+you may (now) geue Cubes, one to the other, in any proportiõ geuẽ: Rationall or Irrationall+: on this maner. Make a hollow Parallelipipedon of Copper or Tinne: with one Base wãting, or open: as in our Cubike Coffen. Frõ the bottome of that Parallelipipedon, raise vp, many perpendiculars, in euery of his fower sides. Now if any proportion be assigned you, in right lines: Cut one of your perpendiculars (or a line equall to it, or lesse then it) likewise: by the 10. of the sixth of Euclide. And those two partes, set in two sundry lines of those perpendiculars (or you may set them both, in one line) making their beginninges, to be, at the base: and so their lengthes to extend vpward. Now, set your hollow Parallelipipedon, vpright, perpendicularly, steadie. Poure in water, handsomly, to the heith of your shorter line. Poure that water, into the hollow Pyramis or Cone. Marke the place of the rising. Settle your hollow Parallelipipedon againe. Poure water into it: vnto the heith of the second line, exactly.

[* Emptying the first.]

Poure that water * duely into the hollow Pyramis or Cone: Marke now againe, where the water cutteth the same line which you marked before. For, there, as the first marked line, is to the second: So shall the two Radicall sides be, one to the other, of any two Cubes: which, in their Soliditie, shall haue the same proportion, which was at the first assigned: were it Rationall or Irrationall.

Thus, in sundry waies you may furnishe your selfe with such straunge and profitable matter: which, long hath bene wished for. And though it be Naturally done and Mechanically: yet hath it a good Demonstration Mathematicall.

[=The demonstrations of this Dubbling of the Cube, and of the rest.=]

Which is this: Alwaies, you haue two Like Pyramids: or two Like Cones, in the proportions assigned: and like Pyramids or Cones, are in proportion, one to the other, in the proportion of their Homologall sides (or lines) tripled. Wherefore, if to the first, and second lines, found in your hollow Pyramis or Cone, you ioyne a third and a fourth, in continuall proportion: that fourth line, shall be to the first, as the greater Pyramis or Cone, is to the lesse: by the 33. of the eleuenth of Euclide. If Pyramis to Pyramis, or Cone to Cone, be double,

[I. D. = * Hereby, helpe your self to become a præcise practiser. And so consider, how, nothing at all, you are hindred (sensibly) by the Conuexitie of the water.=]

then shall * Line to Line, be also double, &c. But, as our first line, is to the second, so is the Radicall side of our Fundamentall Cube, to the Radicall side of the Cube to be made, or to be doubled: and therefore, to those twaine also, a third and a fourth line, in continuall proportion, ioyned: will geue the fourth line in that proportion to the first, as our fourth Pyramidall, or Conike line, was to his first: but that was double, or treble, &c. as the Pyramids or Cones were, one to an other (as we haue proued) therfore, this fourth, shalbe also double or treble to the first, as the Pyramids or Cones were one to an other: But our made Cube, is described of the second in proportion, of the fower proportionall lines:

[= * By the 33. of the eleuenth booke of Euclide.=]

therfore * as the fourth line, is to the first, so is that Cube, to the first Cube: and we haue proued the fourth line, to be to the first, as the Pyramis or Cone, is to the Pyramis or Cone: Wherefore the Cube is to the Cube, as Pyramis is to Pyramis, or Cone is to Cone.

[I. D. = * And your diligence in practise, can so (in waight of water) performe it: Therefore, now, you are able to geue good reason of your whole doing.=]

But we * Suppose Pyramis to Pyramis, or Cone to Cone, to be double or treble. &c. Therfore Cube, is to Cube, double, or treble, &c. Which was to be demonstrated. And of the Parallelipipedõ, it is euidẽt, that the water Solide Parallelipipedons, are one to the other, as their heithes are, seing they haue one base. Wherfore the Pyramids or Cones, made of those water Parallelipipedons, are one to the other, as the lines are (one to the other) betwene which, our proportion was assigned. But the Cubes made of lines, after the proportiõ of the Pyramidal or Conik _homologall_ lines, are one to the other, as the Pyramides or Cones are, one to the other (as we before did proue) therfore, the Cubes made, shalbe one to the other, as the lines assigned, are one to the other: Which was to be demonstrated. Note.

[* _Note this Corollary._]

* This, my Demonstratiõ is more generall, then onely in Square Pyramis or Cone: Consider well. Thus, haue I, both Mathematically and Mechanically, ben very long in wordes: yet (I trust) nothing tedious to them, who, to these thinges, are well affected. And verily I am forced (auoiding prolixitie) to omit sundry such things, easie to be practised: which to the Mathematicien, would be a great Threasure: and to the Mechanicien, no small gaine.

[* The great Commodities following of these new Inuentions.]

* Now may you, +Betwene two lines giuen, finde two middle proportionals, in Continuall proportion: by the hollow Parallelipipedon, and the hollow Pyramis, or Cone.+ Now, any Parallelipipedon rectangle being giuen: thre right lines may be found, proportionall in any proportion assigned, of which, shal be produced a Parallelipipedon, æquall to the Parallelipipedon giuen. Hereof, I noted somwhat, vpon the 36. proposition, of the 11. boke of _Euclide_. Now, all those thinges, which

Vitruuius in his Architecture, specified hable to be done, by dubbling of the Cube: Or, by finding of two middle proportionall lines, betwene two lines giuen, may easely be performed. Now, that Probleme, which I noted vnto you, in the end of my Addition, vpon the 34. of the 11. boke of _Euclide_, is proued possible. Now, may any regular body, be Transformed into an other, &c. Now, any regular body: any Sphere, yea any Mixt Solid: and (that more is) Irregular Solides, may be made (in any proportiõ assigned) like vnto the body, first giuen. Thus, of a _Manneken_, (as the _Dutch_ Painters terme it) in the same _Symmetrie_, may a Giant be made: and that, with any gesture, by the Manneken vsed: and contrarywise. Now, may you, of any Mould, or Modell of a Ship, make one, of the same Mould (in any assigned proportion) bigger or lesser.

[* □]

Now, may you, of any * Gunne, or little peece of ordinaũce, make an other, with the same _Symmetrie_ (in all pointes) as great, and as little, as you will. Marke that: and thinke on it. Infinitely, +may you apply this, so long sought for, and now so easily concluded: and withall, so willingly and frankly communicated to such, as faithfully deale with vertuous studies.+

[Such is the Fruite of the Mathematicall Sciences and Artes.]

Thus, can the Mathematicall minde, deale Speculatiuely in his own Arte: and by good meanes, Mount aboue the cloudes and sterres: And thirdly, he can, by order, Descend, to frame Naturall thinges, to wonderfull vses: and when he list, retire home into his owne Centre: and there, prepare more Meanes, to Ascend or Descend by: and, all, to the glory of God, and our honest delectation in earth.

Although, the Printer, hath looked for this Præface, a day or two, yet could I not bring my pen from the paper, before I had giuen you comfortable warning, and brief instructions, of some of the Commodities, by _Statike_, hable to be reaped: In the rest, I will therfore, be as brief, as it is possible: and with all, describing them, somwhat accordingly. And that, you shall perceiue, by this, which in order commeth next. For, wheras, it is so ample and wonderfull, that, an whole yeare long, one might finde fruitfull matter therin, to speake of: and also in practise, is a Threasure endeles: yet will I glanse ouer it, with wordes very few.

This do I call +‡Anthropographie‡+. Which is an Art restored, and of my preferment to your Seruice. I pray you, thinke of it, as of one of the chief pointes, of Humane knowledge. Although it be, but now, first Cõfirmed, with this new name: yet the matter, hath from the beginning, ben in consideration of all perfect Philosophers. +Anthropographie, is the description of the Number, Measure, Waight, figure, Situation, and colour of euery diuerse thing, conteyned in the perfect body of MAN: with certain knowledge of the Symmetrie, figure, waight, Characterization, and due locall motion, of any parcell of the sayd body, assigned: and of Nũbers, to the sayd parcell appertainyng.+ This, is the one part of the Definition, mete for this place: Sufficient to notifie, the particularitie, and excellency of the Arte: and why it is, here, ascribed to the Mathematicals. Yf the description of the heauenly part of the world, had a peculier Art, called _Astronomie:_ If the description of the earthly Globe, hath

his peculier arte, called _Geographie_. If the Matching of both, hath his peculier Arte, called _Cosmographie:_ Which is the Descriptiõ of the whole, and vniuersall frame of the world: Why should not the description of

[MAN is the Lesse World.]

him, who is the Lesse world: and, frõ the beginning, called _Microcosmus_ (that is. _The Lesse World._) And for whose sake, and seruice, all bodily creatures els, were created: Who, also, participateth with Spirites, and Angels: and is made to the Image and similitude of _God_: haue his peculier Art? and be called the _Arte of Artes_: rather, then, either to want a name, or to haue to base and impropre a name? You must of sundry professions, borow or challenge home, peculier partes hereof: and farder procede: as, God, Nature, Reason and Experience shall informe you. The Anatomistes will restore to you, some part: The Physiognomistes, some: The Chyromantistes some. The Metaposcopistes, some: The excellent, _Albert Durer_, a good part: the Arte of Perspectiue, will somwhat, for the Eye, helpe forward: _Pythagoras_, _Hipocrates_, _Plato_, _Galenus_, _Meletius_, & many other (in certaine thinges) will be Contributaries. And farder, the Heauen, the Earth, and all other Creatures, will eche shew, and offer their Harmonious seruice, to fill vp, that, which wanteth hereof: and with your own Experience, concluding: you may Methodically register the whole, for the posteritie: Whereby, good profe will be had, of our Harmonious, and

[Micro Cosmus.]

Microcosmicall constitution.

[* □]

The outward Image, and vew hereof: to the Art of _Zographie_ and Painting, to Sculpture, and Architecture: (for Church, House, Fort, or Ship) is most necessary and profitable: for that, it is the chiefe base and foundation of them.

[* Lib. 3. Cap. 1.]

Looke in * _Vitruuius_, whether I deale sincerely for your behoufe, or no. Looke in _Albertus Durerus_, _De Symmetria humani Corporis_. Looke in the 27. and 28. Chapters, of the second booke, _De occulta Philosophia_. Consider the _Arke_ of _Noe_. And by that, wade farther. Remember the _Delphicall Oracle NOSCE TEIPSVM_ +_(Knowe thy selfe)_+ so long agoe pronounced: of so many a Philosopher repeated: and of the _Wisest_ attempted: And then, you will perceaue, how long agoe, you haue bene called to the Schole, where this Arte might be learned. Well. I am nothing affrayde, of the disdayne of some such, as thinke Sciences and Artes, to be but Seuen. Perhaps, those Such, may, with ignorance, and shame enough, come short of them Seuen also: and yet neuerthelesse they can not prescribe a certaine number of Artes: and in eche, certaine vnpassable boundes, to God, Nature, and mans Industrie. New Artes, dayly rise vp: and there was no such order taken, that,

[□]

All Artes, should in one age, or in one land, or of one man, be made knowen to the world. Let vs embrace the giftes of God, and wayes to wisedome, in this time of grace, from aboue, continually bestowed on them, who thankefully will receiue them: _Et bonis Omnia Cooperabuntur in bonum._

+‡Trochilike,‡ is that Art Mathematicall, which demonstrateth the properties of all Circular motions, Simple and Compounde.+ And bycause the frute hereof, vulgarly receiued, is in Wheles, it hath the name of _Trochilike:_ as a man would say, _Whele Art_. By this art, a Whele may be geuen which shall moue ones about, in any tyme assigned. Two Wheles may be giuen, whose turnynges about in one and the same tyme, (or equall tymes), shall haue, one to the other, any proportion appointed. By Wheles, may a straight line be described: Likewise, a Spirall line in plaine, Conicall Section lines, and other Irregular lines, at pleasure, may be drawen. These, and such like, are principall Conclusions of this Arte: and helpe forward many pleasant and profitable Mechanicall workes:

[Saw Milles.]

As Milles, to Saw great and very long Deale bordes, no man being by. Such haue I seene in Germany: and in the Citie of Prage: in the kingdome of Bohemia: Coyning Milles, Hand Milles for Corne grinding: And all maner of Milles, and Whele worke: By Winde, Smoke, Water, Waight, Spring, Man or Beast, moued. Take in your hand, _Agricola De re Metallica:_ and then shall you (in all Mines) perceaue, how great nede is, of Whele worke. By Wheles, straunge workes and incredible, are done: as will, in other Artes hereafter, appeare. A wonderfull example of farther possibilitie, and present commoditie, was sene in my time, in a certaine Instrument: which by the Inuenter and Artificer (before) was solde for xx. Talentes of Golde: and then had (by misfortune) receaued some iniurie and hurt: And one _Ianellus_ of _Cremona_ did mend the same, and presented it vnto the Emperour _Charles_ the fifth. _Hieronymus Cardanus_, can be my witnesse, that therein, was one Whele, which moued, and that, in such rate, that, in 7000. yeares onely, his owne periode should be finished. A thing almost incredible: But how farre, I keepe me within my boundes: very many men (yet aliue) can tell.

+‡Helicosophie‡+, is nere Sister to _Trochilike:_ and is, +An Arte Mathematicall, which demonstrateth the designing of all Spirall lines in Plaine, on Cylinder, Cone, Sphære, Conoid, and Sphæroid, and their properties appertayning.+ The vse hereof, in _Architecture_, and diuerse Instrumentes and Engines, is most necessary. For, in many thinges, the Skrue worketh the feate, which, els, could not be performed. By helpe hereof,

[* Atheneus Lib. 5. cap. 8.]

it is * recorded, that, where all the power of the Citie of Syracusa, was not hable to moue a certaine Ship (being on ground) mightie _Archimedes_, setting to, his Skruish Engine, caused _Hiero_ the king, by him self, at ease, to remoue her, as he would.

[Proclus. Pag. 18.]

Wherat, the King wondring: Απὸ τάυτης τῆς ἡμέρας, περὶ παντὸς, Αρχιμήδει λέγοντι πιϛευτεόν. _From this day, forward_ (said the King) _Credit ought to be giuen to Archimedes, what soeuer he sayth._

+‡Pneumatithmie‡ demonstrateth by close hollow Geometricall Figures, (regular and irregular) the straunge properties (in motion or stay) of the Water, Ayre, Smoke, and Fire, in theyr cõtinuitie, and as they are ioyned to the Elementes next them.+ This Arte, to the Naturall Philosopher, is very proffitable: to proue, that _Vacuum_, or _Emptines_ is not in the world. And that, all Nature, abhorreth it so much: that, contrary to ordinary law, the Elementes will moue or stand. As, Water to ascend: rather then betwene him and Ayre, Space or place should be left, more then (naturally) that quãtitie of Ayre requireth, or can fill. Againe, Water to hang, and not descend: rather then by descending, to leaue Emptines at his backe. The like, is of Fire and Ayre: they will descend: when, either, their Cõtinuitie should be dissolued: or their next Element forced from them. And as they will not be extended, to discontinuitie: So, will they not, nor yet of mans force, can be prest or pent, in space, not sufficient and aunswerable to their bodily substance. Great force and violence will they vse, to enioy their naturall right and libertie.

[To go to the bottom of the Sea without daunger.]

Hereupon, two or three men together, by keping Ayre vnder a great Cauldron, and forcyng the same downe, orderly, may without harme descend to the Sea bottome: and continue there a tyme &c. Where, Note, how the thicker Element (as the Water) giueth place to the thynner (as, is the ayre:) and receiueth violence of the thinner, in maner. &c. Pumps and all maner of Bellowes, haue their ground of this Art: and many other straunge deuises. As, _Hydraulica_, Organes goyng by water. &c. Of this Feat, (called commonly _Pneumatica_,) goodly workes are extant, both in Greke, and Latin. With old and learned Schole men, it is called _Scientia de pleno & vacuo._

+‡Menadrie‡, is an Arte Mathematicall, which demonstrateth, how, aboue Natures vertue and power simple: Vertue and force may be multiplied: and so, to direct, to lift, to pull to, and to put or cast fro, any multiplied or simple, determined Vertue, Waight or Force: naturally, not, so, directible or moueable.+ Very much is this Art furdred by other Artes: as, in some pointes, by _Perspectiue_: in some, by _Statike_: in some, by _Trochilike_: and in other, by _Helicosophie_: and _Pneumatithmie_. By this Art, all Cranes, Gybbettes, & Ingines to lift vp, or to force any thing, any maner way, are ordred: and the certaine cause of their force, is knowne: As, the force which one man hath with the Duche waghen Racke: therwith, to set vp agayne, a mighty waghen laden, being ouerthrowne. The force of the Crossebow Racke, is certainly, here, demonstrated. The reason, why one mã, doth with a leauer, lift that, which Sixe men, with their handes onely, could not, so easily do. By this Arte, in our common Cranes in London, where powre is to Crane vp, the waight of 2000. pound: by two Wheles more (by good order added) Arte concludeth, that there may be Craned vp 200000. pound waight &c. So well knew _Archimedes_ this Arte: that he alone, with his deuises and engynes, (twise or thrise) spoyled and discomfited the whole Army and Hoste of the Romaines, besieging _Syracusa_,

[=Plutarchus in Marco Marcello.=]

Marcus Marcellus the Consul, being their Generall Capitaine.

[=Synesius in Epistolis.=]

Such huge Stones, so many, with such force, and so farre, did he with his engynes hayle among them, out of the Citie.

[=Polybius.=]

[=Plinius.=]

[=Quintilianus.=]

[=T. Liuius.=]

And by Sea likewise: though their Ships might come to the walls of _Syracusa_, yet hee vtterly confounded the Romaine Nauye. What with his mighty Stones hurlyng:

[=* Athenæus.=]

what with Pikes of * 18 fote long, made like shaftes: which he forced almost a quarter of a myle. What, with his catchyng hold of their Shyps, and hoysing them vp aboue the water, and suddenly letting them fall into the Sea againe:

[= * Galenus.=]

[=Anthemius.=]

what with his * Burning Glasses: by which he fired their other Shippes a far-of: what, with his other pollicies, deuises, and engines, he so manfully acquit him selfe: that all the Force, courage, and pollicie of the Romaines (for a great season) could nothing preuaile, for the winning of Syracusa. Wherupon, the Romanes named _Archimedes_, _Briareus_, and _Centimanus_. _Zonaras_ maketh mention of one _Proclus_, who so well had perceiued _Archimedes_ Arte of _Menadrie_, and had so well inuented of his owne, that with his Burning Glasses,

[Burning Glasses.]

being placed vpon the walles of Bysance, he multiplied so the heate of the Sunne, and directed the beames of the same against his enemies Nauie with such force, and so sodeinly (like lightening) that he burned and destroyed both man and ship. And _Dion_specifieth of _Priscus_, a _Geometricien_ in Bysance, who inuented and vsed sondry Engins, of Force multiplied: Which was cause, that the _Emperour Seuerus_ pardoned him, his life, after he had wonne Bysance: Bycause he honored the Arte, wytt, and rare industrie of _Priscus_. But nothing inferior to the inuention of these engines of Force, was the inuention of Gunnes.

[Gunnes.]

Which, from an English man, had the occasion and order of first inuenting: though in an other land, and by other men, it was first executed. And they that should see the record, where the occasion and order generall, of Gunning, is first discoursed of, would thinke: that, “small thinges, slight, and cõmon: comming to wise mens

consideration, and industrious mens handling, may grow to be of force incredible.”

+‡Hypogeiodie‡, is an Arte Mathematicall, demonstratyng, how, vnder the Sphæricall Superficies of the earth, at any depth, to any perpendicular line assigned (whose distance from the perpendicular of the entrance: and the Azimuth, likewise, in respect of the said entrance, is knowen) certaine way may be præscribed and gone: And how, any way aboue the Superficies of the earth designed, may vnder earth, at any depth limited, be kept: goyng alwayes, perpendicularly, vnder the way, on earth designed: And, contrarywise, Any way, (straight or croked,) vnder the earth, beyng giuen: vppon the vtface, or Superficies of the earth, to Lyne out the same: So, as, from the Centre of the earth, perpendiculars drawen to the Sphæricall Superficies of the earth, shall precisely fall in the Correspondent pointes of those two wayes. This, with all other Cases and circumstances herein, and appertenances, this Arte demonstrateth.+ This Arte, is very ample in varietie of Conclusions: and very profitable sundry wayes to the Common Wealth. The occasion of my Inuenting this Arte, was at the request of two Gentlemen, who had a certaine worke (of gaine) vnder ground: and their groundes did ioyne ouer the worke: and by reason of the crokednes, diuers depthes, and heithes of the way vnder ground, they were in doubt, and at controuersie, vnder whose ground, as then, the worke was. The name onely (before this) was of me published, _De Itinere Subterraneo_: The rest, be at Gods will. For Pioners, Miners, Diggers for Mettalls, Stone, Cole, and for secrete passages vnder ground, betwene place and place (as this land hath diuerse) and for other purposes, any man may easily perceaue, both the great fruite of this Arte, and also in this Arte, the great aide of Geometrie.

+‡Hydragogie‡, demonstrateth the possible leading of Water, by Natures lawe, and by artificiall helpe, from any head (being a Spring, standing, or running Water) to any other place assigned.+ Long, hath this Arte bene in vse: and much thereof written: and very marueilous workes therein, performed: as may yet appeare, in Italy: by the Ruynes remaining of the Aqueductes. In other places, of Riuers leading through the Maine land, Nauigable many a Mile. And in other places, of the marueilous forcinges of Water to Ascend. which all, declare the great skill, to be required of him, who should in this Arte be perfecte, for all occasions of waters possible leading. To speake of the allowance of the Fall, for euery hundred foote: or of the Ventills (if the waters labour be farre, and great) I neede not: Seing, at hand (about vs) many expert men can sufficiently testifie, in effecte, the order: though the Demonstration of the Necessitie thereof, they know not: Nor yet, if they should be led, vp and downe, and about Mountaines, from the head of the Spring: and then, a place being assigned: and of them, to be demaunded, how low or high, that last place is, in respecte of the head, from which (so crokedly, and vp and downe) they be come: Perhaps, they would not, or could not, very redily, or nerely assoyle that question. _Geometrie_ therefore, is necessary to _Hydragogie_. Of the sundry wayes to force water to ascend, eyther by _Tympane_, _Kettell mills_, _Skrue_, _Ctesibike_, or such like: in _Vitruuius_, _Agricola_, (and other,) fully, the maner may appeare. And so, thereby, also be most euident, how the Artes, of _Pneumatithmie_, _Helicosophie_, _Statike_, _Trochilike_, and _Menadrie_, come to the furniture of this, in Speculation, and to the Commoditie of the Common Wealth, in practise.

+‡Horometrie‡, is an Arte Mathematicall, which demõstrateth, how, at all times appointed, the precise vsuall denominatiõ of time, may be knowen, for any place assigned.+ These wordes, are smoth and plaine easie Englishe, but the reach of their meaning, is farther, then you woulde lightly imagine. Some part of this Arte, was called in olde time, _Gnomonice_: and of late, _Horologiographia_: and in Englishe, may be termed, _Dialling_. Auncient is the vse, and more auncient, is the Inuention. The vse, doth well appeare to haue bene (at the least) aboue two thousand and three hundred yeare agoe:

[4. Reg. 20.]

in * King _Achaz_ Diall, then, by the Sunne, shewing the distinction of time. By Sunne, Mone, and Sterres, this Dialling may be performed, and the precise Time of day or night knowen. But the demonstratiue delineation of these Dialls, of all sortes, requireth good skill, both of _Astronomie_, and _Geometrie_ Elementall, Sphæricall, Phænomenall, and Conikall. Then, to vse the groundes of the Arte, for any regular Superficies, in any place offred: and (in any possible apt position therof) theron, to describe (all maner of wayes) how, vsuall howers, may be (by the _Sunnes_ shadow) truely determined: will be found no sleight Painters worke. So to Paint, and prescribe the Sunnes Motion, to the breadth of a heare. In this Feate (in my youth) I Inuented a way, +How in any Horizontall, Murall, or Æquinoctiall Diall, &c. At all howers (the Sunne shining) the Signe and Degree ascendent, may be knowen.+ Which is a thing very necessary for the Rising of those fixed Sterres: whose Operation in the Ayre, is of great might, euidently. I speake no further, of the vse hereof. Bur forasmuch as, Mans affaires require knowledge of Times & Momentes, when, neither Sunne, Mone, or Sterre, can be sene: Therefore, by Industrie Mechanicall, was inuented, first, how, by Water, running orderly, the Time and howers might be knowen: whereof, the famous _Ctesibius_, was Inuentor: a man, of _Vitruuius_, to the Skie (iustly) extolled. Then, after that, by Sand running, were howers measured: Then, by _Trochilike_ with waight: And of late time, by _Trochilike_ with Spring: without waight. All these, by Sunne or Sterres direction (in certaine time) require ouersight and reformation, according to the heauenly Æquinoctiall Motion: besides the inæqualitie of their owne Operation. There remayneth (without parabolicall meaning herein) among the Philosophers,

[A perpetuall Motion.]

a more excellent, more commodious, and more marueilous way, then all these: of hauing the motion of the Primouant (or first æquinoctiall motion,) by Nature and Arte, Imitated: which you shall (by furder search in waightier studyes) hereafter, vnderstand more of. And so, it is tyme to finish this Annotation, of Tymes distinction, vsed in our common, and priuate affaires: The commoditie wherof, no man would want, that can tell, how to bestow his tyme.

+‡Zographie‡, is an Arte Mathematicall, which teacheth and demonstrateth, how, the Intersection of all visuall Pyramides, made by any playne assigned, (the Centre, distance, and lightes, beyng determined) may be, by lynes, and due propre colours, represented.+ A notable Arte, is this: and would require a whole Volume, to declare

the property thereof: and the Commodities ensuyng. Great skill of _Geometrie_, _Arithmetike_, _Perspectiue_, and _Anthropographie_, with many other particular Artes, hath the _Zographer_, nede of, for his perfection. For, the most excellent Painter, (who is but the propre Mechanicien, & Imitator sensible, of the Zographer) hath atteined to such perfection, that Sense of Man and beast, haue iudged thinges painted, to be things naturall, and not artificiall: aliue, and not dead. This Mechanicall Zographer (commonly called the Painter) is meruailous in his skill: and seemeth to haue a certaine diuine power: As, of frendes absent, to make a frendly, present comfort: yea, and of frendes dead, to giue a continuall, silent presence: not onely with vs, but with our posteritie, for many Ages. And so procedyng, Consider, How, in Winter, he can shew you, the liuely vew of Sommers Ioy, and riches: and in Sommer, exhibite the countenance of Winters dolefull State, and nakednes. Cities, Townes, Fortes, Woodes, Armyes, yea whole Kingdomes (be they neuer so farre, or greate) can he, with ease, bring with him, home (to any mans Iudgement) as Paternes liuely, of the thinges rehearsed. In one little house, can he, enclose (with great pleasure of the beholders,) the portrayture liuely, of all visible Creatures, either on earth, or in the earth, liuing: or in the waters lying, Creping, slyding, or swimming: or of any foule, or fly, in the ayre flying. Nay, in respect of the Starres, the Skie, the Cloudes: yea, in the shew of the very light it selfe (that Diuine Creature) can he match our eyes Iudgement, most nerely. What a thing is this? thinges not yet being, he can represent so, as, at their being, the Picture shall seame (in maner) to haue Created them. To what Artificer, is not Picture, a great pleasure and Commoditie? Which of them all, will refuse the Direction and ayde of Picture? The Architect, the Goldsmith, and the Arras Weauer: of Picture, make great account. Our liuely Herbals, our portraitures of birdes, beastes, and fishes: and our curious Anatomies, which way, are they most perfectly made, or with most pleasure, of vs beholden? Is it not, by Picture onely? And if Picture, by the Industry of the Painter, be thus commodious and meruailous: what shall be thought of _Zographie_, the Scholemaster of Picture, and chief gouernor? Though I mencion not _Sculpture_, in my Table of Artes Mathematicall: yet may all men perceiue, How, that _Picture_ and _Sculpture_, are Sisters germaine: and both, right profitable, in a Commõ wealth. and of _Sculpture_, aswell as of Picture, excellent Artificers haue written great bokes in commendation. Witnesse I take, of _Georgio Vasari_, _Pittore Aretino_: of _Pomponius Gauricus_: and other. To these two Artes, (with other,) is a certaine od Arte, called _Althalmasat_, much beholdyng: more, then the common _Sculptor_, _Entayler_, _Keruer_, _Cutter_, _Grauer_, _Founder_, or _Paynter (&c)_ know their Arte, to be commodious.

[An objection.]

+‡Architecture‡+, to many may seme not worthy, or not mete, to be reckned among the _Artes Mathematicall_. To whom, I thinke good, to giue some account of my so doyng. Not worthy, (will they say,) bycause it is but for building, of a house, Pallace, Church, Forte, or such like, grosse workes. And you, also, defined the _Artes Mathematicall_, to be such, as dealed with no Materiall or corruptible thing: and also did demonstratiuely procede in their faculty, by Number or Magnitude. First,

[The Answer.]

you see, that I count, here, _Architecture_, among those _Artes Mathematicall_, which are Deriued from the Principals: and you know, that such, may deale with Naturall thinges, and sensible matter. Of which, “some draw nerer, to the Simple and absolute Mathematicall Speculation, then other do.

[□]

And though, the _Architect_ procureth, enformeth, & directeth, the _Mechanicien_, to handworke, & the building actuall, of house, Castell, or Pallace, and is chief Iudge of the same: yet, with him selfe (as chief _Master_ and _Architect_,) remaineth the Demonstratiue reason and cause, of the Mechaniciens worke: in Lyne, plaine, and Solid: by _Geometricall_, _Arithmeticall_, _Opticall_, _Musicall_, _Astronomicall_, _Cosmographicall_” (& to be brief) by all the former Deriued _Artes Mathematicall_, and other Naturall Artes, hable to be confirmed and stablished. If this be so: then, may you thinke, that _Architecture_, hath good and due allowance, in this honest Company of _Artes Mathematicall_ Deriuatiue. I will, herein, craue Iudgement of two most perfect _Architectes_: the one, being _Vitruuius_, the Romaine: who did write ten bookes thereof, to the Emperour _Augustus_ (in whose daies our Heauenly Archemaster, was borne): and the other, _Leo Baptista Albertus_, a Florentine: who also published ten bookes therof. _Architectura_ (sayth _Vitruuius_) _est Scientia pluribus disciplinis & varijs eruditionibus ornata: cuius Iudicio probantur omnia, quæ ab cæteris Artificibus perficiuntur opera._ That is. +Architecture, is a Science garnished with many doctrines & diuerse instructions: by whose Iudgement, all workes, by other workmen finished, are Iudged.+ It followeth. _Ea nascitur ex Fabrica, & Ratiocinatione. &c. Ratiocinatio autem est, quæ, res fabricatas, Solertia ac ratione proportionis, demonstrare at[que] explicare potest. +Architecture, groweth of Framing, and Reasoning. &c. Reasoning, is that, which of thinges framed, with forecast, and proportion: can make demonstration, and manifest declaration.+_ Againe. _Cùm, in omnibus enim rebus, tùm maximè etiam in Architectura, hæc duo insunt: quod significatur, & quod significat. Significatur proposita res, de qua dicitur: hanc autem Significat Demonstratio, rationibus doctrinarum explicata. +Forasmuch as, in all thinges: therefore chiefly in Architecture, these two thinges are: the thing signified: and that which signifieth. The thing propounded, whereof we speake, is the thing Signified. But Demonstration, expressed with the reasons of diuerse doctrines, doth signifie the same thing.+_ After that. _Vt literatus sit, peritus Graphidos, eruditus Geometriæ, & Optices non ignarus: instructus Arithmetica: historias complures nouerit, Philosophos diligenter audiuerit: Musicam sciuerit: Medicinæ non sit ignarus, responsa Iurisperitorũ nouerit: Astrologiam, Cæli[que] rationes cognitas habeat. +An Architect+_ (sayth he) +_ought to vnderstand Languages, to be skilfull of Painting, well instructed in Geometrie, not ignorant of Perspectiue, furnished with Arithmetike, haue knowledge of many histories, and diligently haue heard Philosophers, haue skill of Musike, not ignorant of Physike, know the aunsweres of Lawyers, and haue Astronomie, and the courses Cælestiall, in good knowledge._+ He geueth reason, orderly, wherefore all these Artes, Doctrines, and Instructions, are requisite in an excellent _Architect_. And (for breuitie) omitting the Latin text, thus he hath. +_Secondly, it is behofefull for an Architect to haue the knowledge of Painting: that he may the more easilie fashion out, in patternes painted, the forme of

what worke he liketh. And Geometrie, geueth to Architecture many helpes: and first teacheth the Vse of the Rule, and the Cumpasse: wherby (chiefly and easilie) the descriptions of Buildinges, are despatched in Groundplats: and the directions of Squires, Leuells, and Lines. Likewise, by Perspectiue, the Lightes of the heauen, are well led, in the buildinges: from certaine quarters of the world. By Arithmetike, the charges of Buildinges are summed together: the measures are expressed, and the hard questions of Symmetries, are by Geometricall Meanes and Methods discoursed on. &c. Besides this, of the Nature of thinges (which in Greke is called φυσιολογία) Philosophie doth make declaration. Which, it is necessary, for an Architect, with diligence to haue learned: because it hath many and diuers naturall questions: as specially, in Aqueductes. For in their courses, leadinges about, in the leuell ground, and in the mountinges, the naturall Spirites or breathes are ingendred diuers wayes: The hindrances, which they cause, no man can helpe, but he, which out of Philosophie, hath learned the originall causes of thinges. Likewise, who soeuer shall read Ctesibius, or Archimedes bookes, (and of others, who haue written such Rules) can not thinke, as they do: vnlesse he shall haue receaued of Philosophers, instructions in these thinges. And Musike he must nedes know: that he may haue vnderstanding, both of Regular and Mathematicall Musike: that he may temper well his Balistes, Catapultes, and Scorpions. &c. Moreouer, the Brasen Vessels, which in Theatres, are placed by Mathematicall order, in ambries, vnder the steppes: and the diuersities of the soundes (which ye Grecians call ηχεῖα) are ordred according to Musicall Symphonies & Harmonies: being distributed in ye Circuites, by Diatessaron, Diapente, and Diapason. That the conuenient voyce, of the players sound, whẽ it came to these preparations, made in order, there being increased: with y^t increasing, might come more cleare & pleasant, to ye eares of the lokers on. &c. And of Astronomie, is knowẽ ye East, West, South, and North. The fashion of the heauen, the Æquinox, the Solsticie, and the course of the sterres. Which thinges, vnleast one know: he can not perceiue, any thyng at all, the reason of Horologies. Seyng therfore this ample Science, is garnished, beautified and stored, with so many and sundry skils and knowledges: I thinke, that none can iustly account them selues Architectes, of the suddeyne. But they onely, who from their childes yeares, ascendyng by these degrees of knowledges, beyng fostered vp with the atteynyng of many Languages and Artes, haue wonne to the high Tabernacle of Architecture. &c. And to whom Nature hath giuen such quicke Circumspection, sharpnes of witt, and Memorie, that they may be very absolutely skillfull in Geometrie, Astronomie, Musike, and the rest of the Artes Mathematicall: Such, surmount and passe the callyng, and state, of Architectes:

[A Mathematicien.]

and are become Mathematiciens. &c. And they are found, seldome. As, in tymes past, was Aristarchus Samius: Philolaus, and Archytas, Tarentynes: Apollonius Pergęus: Eratosthenes Cyreneus: Archimedes, and Scopas, Syracusians. Who also, left to theyr posteritie, many Engines and Gnomonicall workes: by numbers and naturall meanes, inuented and declared._+

Thus much, and the same wordes (in sense) in one onely Chapter of this Incõparable _Architect Vitruuius_, shall you finde. And if you should, but take his boke in your hand, and slightly loke thorough it, you would say straight way:

[Vitruuius.]

This is _Geometrie_, _Arithmetike_, _Astronomie_, _Musike_, _Anthropographie_, _Hydragogie_, _Horometrie_. _&c_. and (to cõclude) the Storehouse of all workmãship. Now, let vs listen to our other Iudge, our Florentine, _Leo Baptista_: and narrowly consider, how he doth determine of _Architecture_. _Sed ante[que] vltra progrediar. &c. +But before I procede any further +_(sayth he) +_I thinke, that I ought to expresse, what man I would haue to bee allowed an Architect. For, I will not bryng in place a Carpenter: as though you might Compare him to the Chief Masters of other Artes. For the hand of the Carpenter, is the Architectes Instrument._+

[VVho is an Architect.]

+_But I will appoint the Architect to be “that man, who hath the skill, (by a certaine and meruailous meanes and way,) both in minde and Imagination to determine and also in worke to finish: what workes so euer, by motion of waight, and cuppling and framyng together of bodyes, may most aptly be Commodious for the worthiest Vses of Man.” And that he may be able to performe these thinges, he hath nede of atteynyng and knowledge of the best, and most worthy thynges. &c. The whole Feate of Architecture in buildyng, consisteth in Lineamentes, and in Framyng. And the whole power and skill of Lineamentes, tendeth to this: that the right and absolute way may be had, of Coaptyng and ioyning Lines and angles: by which, the face of the buildyng or frame, may be comprehended and concluded. And it is the property of Lineamentes, to prescribe vnto buildynges, and euery part of them, an apt place, & certaine nũber: a worthy maner, and a semely order: that, so, ye whole forme and figure of the buildyng, may rest in the very Lineamentes. &c. And we may prescribe in mynde and imagination the whole formes, *

[* The Immaterialitie of perfect Architecture.]

all material stuffe beyng secluded. Which point we shall atteyne, by Notyng and forepointyng the angles, and lines, by a sure and certaine direction and connexion. Seyng then, these thinges, are thus:_+

[What, Lineament is.]

+_Lineamente, shalbe the certaine and constant prescribyng, conceiued in mynde: made in lines and angles: and finished with a learned minde and wyt._+ “We thanke you Master _Baptist_, that you haue so aptly brought your Arte, and phrase therof, to haue some Mathematicall perfection:

[Note.]

by certaine order, nũber, forme, figure, and _Symmetrie_ mentall:” all naturall & sensible stuffe set a part. Now, then, it is euident, (Gentle reader) how aptely and worthely, I haue preferred _Architecture_, to be bred and fostered vp in the Dominion of the pereles _Princesse_, _Mathematica_: and to be a naturall Subiect of hers. And

the name of _Architecture_, is of the principalitie, which this Science hath, aboue all other Artes. And _Plato_ affirmeth, the _Architect_ to be _Master_ ouer all, that make any worke. Wherupon, he is neither Smith, nor Builder: nor, separately, any Artificer: but the Hed, the Prouost, the Directer, and Iudge of all Artificiall workes, and all Artificers. For, the true _Architect_, is hable to teach, Demonstrate, distribute, describe, and Iudge all workes wrought. And he, onely, searcheth out the causes and reasons of all Artificiall thynges. Thus excellent, is _Architecture_: though few (in our dayes) atteyne thereto: yet may not the Arte, be otherwise thought on, then in very dede it is worthy. Nor we may not, of auncient Artes, make new and imperfect Definitions in our dayes: for scarsitie of Artificers: No more, than we may pynche in, the Definitions of _Wisedome_, or _Honestie_, or of _Frendeshyp_ or of _Iustice_. No more will I consent, to Diminish any whit, of the perfection and dignitie, (by iust cause) allowed to absolute _Architecture_. Vnder the Direction of this Arte, are thre principall, necessary _Mechanicall Artes_. Namely, _Howsing_, _Fortification_, and _Naupegie_. _Howsing_, I vnderstand, both for Diuine Seruice, and Mans common vsage: publike, and priuate. Of _Fortification_ and _Naupegie_, straunge matter might be told you: But perchaunce, some will be tyred, with this Bederoll, all ready rehearsed: and other some, will nycely nip my grosse and homely discoursing with you: made in post hast: for feare you should wante this true and frendly warnyng, and tast giuyng, of the _Power Mathematicall_. Lyfe is short, and vncertaine: Tymes are perilouse: &c. And still the Printer awayting, for my pen staying: All these thinges, with farder matter of Ingratefulnes, giue me occasion to passe away, to the other Artes remainyng, with all spede possible.

+The Arte of ‡Nauigation‡, demonstrateth how, by the shortest good way, by the aptest Directiõ, & in the shortest time, a sufficient Ship, betwene any two places (in passage Nauigable,) assigned: may be cõducted: and in all stormes, & naturall disturbances chauncyng, how, to vse the best possible meanes, whereby to recouer the place first assigned.+ What nede, the _Master Pilote_, hath of other Artes, here before recited, it is easie to know: as, of _Hydrographie_, _Astronomie_, _Astrologie_, and _Horometrie_. Presupposing continually, the common Base, and foundacion of all: namely _Arithmetike_ and _Geometrie_. So that, he be hable to vnderstand, and Iudge his own necessary Instrumentes, and furniture Necessary: Whether they be perfectly made or no: and also can, (if nede be) make them, hym selfe. As Quadrantes, The Astronomers Ryng, The Astronomers staffe, The Astrolabe vniuersall. An Hydrographicall Globe. Charts Hydrographicall, true, (not with parallell Meridians). The Common Sea Compas: The Compas of variacion: The Proportionall, and Paradoxall Compasses

[Anno. 1559.]

(of me Inuented, for our two Moscouy Master Pilotes, at the request of the Company) Clockes with spryng: houre, halfe houre, and three houre Sandglasses: & sundry other Instrumẽtes: And also, be hable, on Globe, or Playne to describe the Paradoxall Compasse: and duely to vse the same, to all maner of purposes, whereto it was inuented. And also, be hable to Calculate the Planetes places for all tymes.

Moreouer, with Sonne Mone or Sterre (or without) be hable to define the Longitude & Latitude of the place, which he is in: So that, the Longitude & Latitude of the place, from which he sayled, be giuen: or by him, be knowne. whereto, appertayneth expert meanes, to be certified euer, of the Ships way. &c. And by foreseing the Rising, Settyng, Nonestedyng, or Midnightyng of certaine tempestuous fixed Sterres: or their Coniunctions, and Anglynges with the Planetes, &c. he ought to haue expert coniecture of Stormes, Tempestes, and Spoutes: and such lyke Meteorologicall effectes, daungerous on Sea. For (as _Plato_ sayth,) _Mutationes, opportunitates[que] temporum presentire, non minus rei militari, quàm Agriculturæ, Nauigationi[que] conuenit. +To foresee the alterations and opportunities of tymes, is conuenient, no lesse to the Art of Warre, then to Husbandry and Nauigation.+_ And besides such cunnyng meanes, more euident tokens in Sonne and Mone, ought of hym to be knowen: such as (the Philosophicall Poëte) _Virgilius_ teacheth, in hys _Georgikes_. Where he sayth,

[Sidenote: Georgic. 1.]

Sol quo[que] & exoriens & quum se condet in vndas, Signa dabit, Solem certissima signa sequuntur. &c. -------- Nam sæpe videmus, Ipsius in vultu varios errare colores. Cæruleus, pluuiam denunciat, igneus Euros. Sin maculæ incipient rutilo immiscerier igni, Omnia tum pariter vento, nimbis[que] videbis Feruere: non illa quisquam me nocte per altum Ire, ne[que] a terra moueat conuellere funem. &c. Sol tibi signa dabit. Solem quis dicere falsum Audeat? -------- &c.

And so of Mone, Sterres, Water, Ayre, Fire, Wood, Stones, Birdes, and Beastes, and of many thynges els, a certaine Sympathicall forewarnyng may be had: sometymes to great pleasure and proffit, both on Sea and Land. Sufficiently, for my present purpose, it doth appeare, by the premisses, how _Mathematicall_, the _Arte_ of _Nauigation_, is: and how it nedeth and also vseth other _Mathematicall Artes_: And now, if I would go about to speake of the manifold Commodities, commyng to this Land, and others, by Shypps and _Nauigation_, you might thinke, that I catch at occasions, to vse many wordes, where no nede is.

Yet, this one thyng may I, (iustly) say. In _Nauigation_, none ought to haue greater care, to be skillfull, then our English Pylotes. And perchaunce, Some, would more attempt: And other Some, more willingly would be aydyng, it they wist certainely, What Priuiledge, God had endued this Iland with, by reason of Situation, most commodious for _Nauigation_, to Places most Famous & Riche. And though,

[* Anno. 1567 S. H. G.]

(of * Late) a young Gentleman, a Courragious Capitaine, was in a great readynes, with good hope, and great causes of persuasion, to haue ventured, for a Discouerye, (either _Westerly_, by _Cape de Paramantia_: or _Esterly_, aboue _Noua Zemla_, and the _Cyremisses_) and was, at the very nere tyme of Attemptyng, called and employed otherwise (both then, and since,) in great good seruice to his Countrey, as the Irish Rebels haue * tasted:

[* Anno. 1569]

Yet, I say, (though the same Gentleman, doo not hereafter, deale therewith) Some one, or other, should listen to the Matter: and by good aduise, and discrete Circumspection, by little, and little, wynne to the sufficient knowledge of that +Trade+ and +Voyage+: Which, now, I would be sory, (through Carelesnesse, want of Skill, and Courrage,) should remayne Vnknowne and vnheard of. Seyng, also, we are herein, halfe Challenged, by the learned, by halfe request, published. Therof, verely, might grow Commoditye, to this Land chiefly, and to the rest of the Christen Common wealth, farre passing all riches and worldly Threasure.

+‡Thaumaturgike‡, is that Art Mathematicall, which giueth certaine order to make straunge workes, of the sense to be perceiued, and of men greatly to be wondred at.+ By sundry meanes, this _Wonder-worke_ is wrought. Some, by _Pneumatithmie_. As the workes of _Ctesibius_ and _Hero_, Some by waight. wherof _Timæus_ speaketh. Some, by Stringes strayned, or Springs, therwith Imitating liuely Motions. Some, by other meanes, as the Images of Mercurie: and the brasen hed, made by _Albertus Magnus_, which dyd seme to speake. _Boethius_ was excellent in these feates. To whom, _Cassiodorus_ writyng, sayth. +_Your purpose is to know profound thynges: and to shew meruayles. By the disposition of your Arte, Metals do low: Diomedes of brasse, doth blow a Trumpet loude: a brasen Serpent hisseth: byrdes made, sing swetely. Small thynges we rehearse of you, who can Imitate the heauen. &c._+ Of the straunge Selfmouyng, which, at Saint Denys, by Paris,

[* Anno. 1551]

* I saw, ones or twise (_Orontius_ beyng then with me, in Company) it were to straunge to tell. But some haue written it. And yet, (I hope) it is there, of other to be sene. And by _Perspectiue_ also straunge thinges, are done. As partly (before) I gaue you to vnderstand in _Perspectiue_. As, to see in the Ayre, a loft, the lyuely Image of an other man, either walkyng to and fro: or standyng still. Likewise, to come into an house, and there to see the liuely shew of Gold, Siluer or precious stones: and commyng to take them in your hand, to finde nought but Ayre. Hereby, haue some men (in all other matters counted wise) fouly ouershot thẽ selues: misdeaming of the meanes. Therfore sayd _Claudius Cælestinus_.

[De his quæ Mundo mirabiliter eueniunt. cap. 8.]

Hodie magnæ literaturæ viros & magna reputationis videmus, opera quedam quasi miranda, supra Naturã putare: de quibus in Perspectiua doctus causam faciliter reddidisset. That is. +_Now a dayes, we see some men, yea of great learnyng and reputation, to Iudge certain workes as meruaylous, aboue the power of Nature: Of which workes, one that were skillfull in Perspectiue might easely haue giuen the Cause._+ Of _Archimedes Sphære_, _Cicero_ witnesseth.

[Tusc. 1.]

Which is very straunge to thinke on. +_For when Archimedes_+ (sayth he) +_did fasten in a Sphære, the mouynges of the Sonne, Mone, and of the fiue other Planets, he did, as the God, which (in Timæus of Plato) did make the world. That, one turnyng, should rule motions most vnlike in slownes, and swiftnes._+ But a greater cause of meruayling we haue by _Claudianus_ report hereof. Who affirmeth this _Archimedes

worke_, to haue ben of Glasse. And discourseth of it more at large: which I omit. The Doue of wood, which the _Mathematicien Archytas_ did make to flye, is by _Agellius_ spoken of. Of _Dædalus_ straunge Images, _Plato_ reporteth. _Homere_ of _Vulcans Selfmouers_, (by secret wheles) leaueth in writyng. _Aristotle_, in hys _Politikes_, of both, maketh mention. Meruaylous was the workemanshyp, of late dayes, performed by good skill of _Trochilike. &c._ For in Noremberge, A flye of Iern, beyng let out of the Artificers hand, did (as it were) fly about by the gestes, at the table, and at length, as though it were weary, retourne to his masters hand agayne. Moreouer, an Artificiall Egle, was ordred, to fly out of the same Towne, a mighty way, and that a loft in the Ayre, toward the Emperour comming thether: and followed hym, beyng come to the gate of the towne. *

[* □]

Thus, you see, what, Arte Mathematicall can performe, when Skill, will, Industry, and Hability, are duely applyed to profe.

[A Digression.]

And for these, and such like marueilous Actes and Feates, Naturally, Mathematically, and Mechanically, wrought and contriued:

[Apologeticall.]

ought any honest Student, and Modest Christian Philosopher, be counted, & called a +Coniurer+? Shall the folly of Idiotes, and the Mallice of the Scornfull, so much preuaile, that He, who seeketh no worldly gaine or glory at their handes: But onely, of God, the threasor of heauenly wisedome, & knowledge of pure veritie: Shall he (I say) in the meane space, be robbed and spoiled of his honest name and fame? He that seketh (by S. Paules aduertisement) in the Creatures Properties, and wonderfull vertues, to finde iuste cause, to glorifie the Æternall, and Almightie Creator by: Shall that man, be (in hugger mugger) condemned, as a Companion of the Helhoundes, and a Caller, and Coniurer of wicked and damned Spirites? He that bewaileth his great want of time, sufficient (to his contentation) for learning of Godly wisdome, and Godly Verities in: and onely therin setteth all his delight: Will that mã leese and abuse his time, in dealing with the Chiefe enemie of Christ our Redemer: the deadly foe of all mankinde: the subtile and impudent peruerter of Godly Veritie: the Hypocriticall Crocodile: the Enuious Basiliske, continually desirous, in the twinke of an eye, to destroy all Mankinde, both in Body and Soule, æternally? Surely (for my part, somewhat to say herein) I haue not learned to make so brutish, and so wicked a Bargaine. Should I, for my xx. or xxv. yeares Studie: for two or three thousand Markes spending: seuen or eight thousand Miles going and trauailing, onely for good learninges sake: And that, in all maner of wethers: in all maner of waies and passages: both early and late: in daunger of violence by man: in daunger of destruction by wilde beastes: in hunger: in thirst: in perilous heates by day, with toyle on foote: in daungerous dampes of colde, by night, almost bereuing life: (as God knoweth): with lodginges, oft times, to small ease: and somtime to lesse securitie. And for much more (then all this) done & suffred, for Learning and attaining of Wisedome: Should I (I pray you) for all this, no otherwise, nor more warily: or (by

Gods mercifulnes) no more luckily, haue fished, with so large, and costly, a Nette, so long time in drawing (and that with the helpe and aduise of Lady Philosophie, & Queene Theologie): but at length, to haue catched, and drawen vp, * a Frog?

[* A prouerb. Fayre fisht, and caught a Frog.]

Nay, a Deuill? For, so, doth the Common peuish Pratler Imagine and Iangle: And, so, doth the Malicious skorner, secretly wishe, & brauely and boldly face down, behinde my backe. Ah, what a miserable thing, is this kinde of Men? How great is the blindnes & boldnes, of the Multitude, in thinges aboue their Capacitie? What a Land: what a People: what Maners: what Times are these? Are they become Deuils, them selues: and, by false witnesse bearing against their Neighbour, would they also, become Murderers? Doth God, so long geue them respite, to reclaime them selues in, from this horrible slaundering of the giltlesse: contrary to their owne Consciences: and yet will they not cease? Doth the Innocent, forbeare the calling of them, Iuridically to aunswere him, according to the rigour of the Lawes: and will they despise his Charitable pacience? As they, against him, by name, do forge, fable, rage, and raise slaunder, by Worde & Print: Will they prouoke him, by worde and Print, likewise, to Note their Names to the World: with their particular deuises, fables, beastly Imaginations, and vnchristen-like slaunders? Well: Well. O (you such) my vnkinde Countrey men. O vnnaturall Countrey men. O vnthankfull Countrey men. O Brainsicke, Rashe, Spitefull, and Disdainfull Countrey men. Why oppresse you me, thus violently, with your slaundering of me: Contrary to Veritie: and contrary to your owne Consciences? And I, to this hower, neither by worde, deede, or thought, haue bene, any way, hurtfull, damageable, or iniurious to you, or yours? Haue I, so long, so dearly, so farre, so carefully, so painfully, so daungerously sought & trauailed for the learning of Wisedome, & atteyning of Vertue: And in the end (in your iudgemẽt) am I become, worse, then when I begã? Worse, thẽ a Mad man? A dangerous Member in the Common Wealth: and no Member of the Church of Christ? Call you this, to be Learned? Call you this, to be a Philosopher? and a louer of Wisedome? To forsake the straight heauenly way: and to wallow in the broad way of damnation? To forsake the light of heauenly Wisedome: and to lurke in the dungeon of the Prince of darkenesse? To forsake the Veritie of God, & his Creatures: and to fawne vpon the Impudent, Craftie, Obstinate Lier, and continuall disgracer of Gods Veritie, to the vttermost of his power? To forsake the Life & Blisse Æternall: and to cleaue vnto the Author of Death euerlasting? that Murderous Tyrant, most gredily awaiting the Pray of Mans Soule? Well: I thanke God and our Lorde Iesus Christ, for the Comfort which I haue by the Examples of other men, before my time: To whom, neither in godlines of life, nor in perfection of learning, I am worthy to be compared: and yet, they sustained the very like Iniuries, that I do: or rather, greater. Pacient _Socrates_, his _Apologie_ will testifie: _Apuleius_ his _Apologies_, will declare the Brutishnesse of the Multitude. _Ioannes Picus_, Earle of Mirandula, his _Apologie_ will teach you, of the Raging slaunder of the Malicious Ignorant against him. _Ioannes Trithemius_, his _Apologie_ will specifie, how he had occasion to make publike Protestation: as well by reason of the Rude Simple: as also, in respect of such, as were counted to be of the wisest sort of men. “Many could I recite: But I deferre the precise and determined handling of this matter: being loth to detect the Folly & Mallice of my Natiue Countrey men. *

[* □]

Who, so hardly, can disgest or like any extraordinary course of Philosophicall Studies: not falling within the Cumpasse of their Capacitie: or where they are not made priuie of the true and secrete cause, of such wonderfull Philosophicall Feates." These men, are of fower sortes, chiefly. The first, I may name, _Vaine pratling busie bodies_: The second, _Fond Frendes_: The third, _Imperfectly zelous_: and the fourth, _Malicious Ignorant_. To eche of these (briefly, and in charitie) I will say a word or two, and so returne to my Præface.

[1.]

Vaine pratling busie bodies, vse your idle assemblies, and conferences, otherwise, then in talke of matter, either aboue your Capacities, for hardnesse: or contrary to your Consciences, in Veritie.

[2.]

Fonde Frendes, leaue of, so to commend your vnacquainted frend, vpon blinde affection: As, because he knoweth more, then the common Student: that, therfore, he must needes be skilfull, and a doer, in such matter and maner, as you terme _Coniuring_. Weening, thereby, you aduaunce his fame: and that you make other men, great marueilers of your hap, to haue such a learned frend. Cease to ascribe Impietie, where you pretend Amitie. For, if your tounges were true, then were that your frend, _Vntrue_, both to God, and his Soueraigne. Such _Frendes_ and _Fondlinges_, I shake of, and renounce you: Shake you of, your Folly.

[3.]

Imperfectly zelous, to you, do I say: that (perhaps) well, do you Meane: But farre you misse the Marke: If a Lambe you will kill, to feede the flocke with his bloud. Sheepe, with Lambes bloud, haue no naturall sustenaunce: No more, is Christes flocke, with horrible slaunders, duely ædified. Nor your faire pretense, by such rashe ragged Rhetorike, any whit, well graced. But such, as so vse me, will finde a fowle Cracke in their Credite. Speake that you know: And know, as you ought: Know not, by Heare say, when life lieth in daunger. Search to the quicke, & let Charitie be your guide.

[4.]

Malicious Ignorant, what shall I say to thee? _Prohibe linguam tuam a malo. A detractione parcite linguæ. +Cause thy toung to refraine frõ euill. Refraine your toung from slaunder.+_ Though your tounges be sharpned, Serpent like, & Adders poyson lye in your lippes:

[Psal. 140.]

yet take heede, and thinke, betimes, with your selfe, _Vir linguosus non stabilietur in terra. Virum violentum venabitur malum, donec præcipitetur._ For, sure I am, _Quia faciet Dominus Iudicium afflicti: & vindictam pauperum._

Thus, I require you, my assured frendes, and Countrey men (you Mathematiciens, Mechaniciens, and Philosophers, Charitable and discrete) to deale in my behalf, with the light & vntrue tounged, my enuious Aduersaries, or Fond frends. And farther, I would wishe, that at leysor, you would consider, how _Basilius Magnus_, layeth _Moses_ and _Daniel_, before the eyes of those, which count all such Studies Philosophicall (as mine hath bene) to be vngodly, or vnprofitable. Waye well _S. Stephen_ his witnesse of _Moses_.

[Act. 7. C.]

Eruditus est Moses omni Sapientia Ægyptiorũ: & erat potens in verbis & operibus suis. +Moses was instructed in all maner of wisedome of the Ægyptians: and he was of power both in his wordes, and workes.+ You see this Philosophicall Power & Wisedome, which _Moses_ had, to be nothing misliked of the Holy Ghost. Yet _Plinius_ hath recorded, _Moses_ to be a wicked _Magicien_. And that (of force) must be, either for this Philosophicall wisedome, learned, before his calling to the leading of the Children of _Israel_: or for those his wonders, wrought before King _Pharao_, after he had the conducting of the _Israelites_. As concerning the first, you perceaue, how _S. Stephen_, at his Martyrdome (being full of the Holy Ghost) in his Recapitulation of the olde Testament, hath made mention of _Moses_ Philosophie: with good liking of it: And _Basilius Magnus_ also, auoucheth it, to haue bene to _Moses_ profitable (and therefore, I say, to the Church of God, necessary). But as cõcerning _Moses_ wonders, done before King _Pharao_: God, him selfe, sayd: _Vide vt omnia ostenta, quæ posui in manu tua, facias coram Pharaone. +See that thou do all those wonders before Pharao, which I haue put in thy hand.+_ Thus, you euidently perceaue, how rashly, _Plinius_ hath slaundered _Moses_,

[Lib. 30. Cap. 1.]

of vayne fraudulent _Magike_, saying: _Est & alia Magices Factio, a Mose, Iamne, & Iotape, Iudæis pendens: sed multis millibus annorum post Zoroastrem. &c._

[1.]

Let all such, therefore, who, in Iudgement and Skill of Philosophie, are farre Inferior to _Plinie_, “take good heede, least they ouershoote them selues rashly,” in

[□]

Iudging of _Philosophers straunge Actes_: and the Meanes, how they are done.

[2.]

But, much more, ought they to beware of forging, deuising, and imagining monstrous feates, and wonderfull workes, when and where, no such were done: no, not any sparke or likelihode, of such, as they, without all shame, do report.

[3.]

And (to conclude) most of all, let them be ashamed of Man, and afraide of the dreadfull and Iuste Iudge: both Folishly or Maliciously to deuise: and then, deuilishly to father their new fond Monsters on me: Innocent, in hand and hart: for trespacing

either against the lawe of God, or Man, in any my Studies or Exercises, Philosophicall, or Mathematicall: As in due time, I hope, will be more manifest.

Now end I, with +‡Archemastrie‡+. Which name, is not so new, as this Arte is rare. For an other Arte, vnder this, a degree (for skill and power) hath bene indued with this English name before. And yet, this, may serue for our purpose, sufficiently, at this present. +This Arte, teacheth to bryng to actuall experience sensible, all worthy conclusions by all the Artes Mathematicall purposed, & by true Naturall Philosophie concluded: & both addeth to them a farder scope, in the termes of the same Artes, & also by hys propre Method, and in peculier termes, procedeth, with helpe of the foresayd Artes, to the performance of complet Experiẽces, which of no particular Art, are hable (Formally) to be challenged.+ If you remember, how we considered _Architecture_, in respect of all common handworkes: some light may you haue, therby, to vnderstand the Souerainty and propertie of this Science. _Science_ I may call it, rather, then an Arte: for the excellency and Mastershyp it hath, ouer so many, and so mighty Artes and Sciences. And bycause it procedeth by _Experiences_, and searcheth forth the causes of Conclusions, by _Experiences_: and also putteth the Conclusions them selues, in _Experience_, it is named of some, _Scientia Experimentalis_. The +_Experimentall Science_+. _Nicolaus Cusanus_ termeth it so, in hys _Experimentes Statikall_, And an other _Philosopher_,

[R. B.]

of this land Natiue (the floure of whose worthy fame, can neuer dye nor wither) did write therof largely, at the request of _Clement the sixt_. The Arte carrieth with it, a wonderfull Credit: By reason, it certefieth, sensibly, fully, and completely to the vtmost power of Nature, and Arte. This Arte, certifieth by _Experience_ complete and absolute: and other Artes, with their Argumentes, and Demonstrations, persuade: and in wordes, proue very well their Conclusions. *

[□]

But wordes, and Argumentes, are no sensible certifying: nor the full and finall frute of Sciences practisable. And though some Artes, haue in them, _Experiences_, yet they are not complete, and brought to the vttermost, they may be stretched vnto, and applyed sensibly. As for example: the Naturall Philosopher disputeth and maketh goodly shew of reason: And the Astronomer, and the Opticall Mechanicien, put some thynges in _Experience_: but neither, all, that they may: nor yet sufficiently, and to the vtmost, those, which they do, There, then, the _Archemaster_ steppeth in, and leadeth forth on, the _Experiences_, by order of his doctrine _Experimentall_, to the chief and finall power of Naturall and Mathematicall Artes. Of two or three men, in whom, this Description of _Archemastry_ was _Experimentally_, verified, I haue read and hard: and good record, is of their such perfection. So that, this Art, is no fantasticall Imagination: as some Sophister, might, _Cum suis Insolubilibus_, make a florish: and dassell your Imagination: and dash your honest desire and Courage, from beleuing these thinges, so vnheard of, so meruaylous, & of such Importance. Well: as you will. I haue forewarned you. I haue done the part of a frende: I haue discharged

my Duety toward God: for my small Talent, at hys most mercyfull handes receiued. To this Science, doth the _Science Alnirangiat_, great Seruice. Muse nothyng of this name. I chaunge not the name, so vsed, and in Print published by other: beyng a name, propre to the Science. Vnder this, commeth _Ars Sintrillia_, by _Artephius_, briefly written. But the chief Science, of the Archemaster, (in this world) as yet knowen, is an other (as it were) OPTICAL Science: wherof, the name shall be told (God willyng) when I shall haue some, (more iust) occasion, therof, to Discourse.

Here, I must end, thus abruptly (Gentle frende, and vnfayned louer of honest and necessary verities.) For, they, who haue (for your sake, and vertues cause) requested me, (an old forworne Mathematicien) to take pen in hand: (through the confidence they reposed in my long experience: and tryed sincerity) for the declaryng and reportyng somewhat, of the frute and commodity, by the +Artes Mathematicall, to be atteyned vnto+: euen they, Sore agaynst their willes, are forced, for sundry causes, to satisfie the workemans request, in endyng forthwith: He, so feareth this, so new an attempt, & so costly: And in matter so slenderly (hetherto) among the common Sorte of Studentes, considered or estemed.

And where I was willed, somewhat to alledge, why, in our vulgare Speche, this part of the Principall Science of _Geometrie_, called _Euclides Geometricall Elementes_, is published, to your handlyng: being vnlatined people, and not Vniuersitie Scholers: Verily, I thinke it nedelesse.

[1.]

For, the Honour, and Estimation of the +Vniuersities, and Graduates+, is, hereby, nothing diminished. Seing, from, and by their Nurse Children, you receaue all this Benefite: how great soeuer it be.

[2.]

Neither are their Studies, hereby, any whit hindred. No more, then the Italian _Vniuersities_, as _Academia Bononiensis_, _Ferrariensis_, _Florentina_, _Mediolanensis_, _Patauina_, _Papiensis_, _Perusina_, _Pisana_, _Romana_, _Senensis_, or any one of them, finde them selues, any deale, disgraced, or their Studies any thing hindred, by _Frater Lucas de Burgo_, or by _Nicolaus Tartalea_, who in vulgar Italian language, haue published, not onely _Euclides Geometrie_, but of _Archimedes_ somewhat: and in Arithmetike and Practicall Geometrie, very large volumes, all in their vulgar speche. Nor in Germany haue the famous _Vniuersities_, any thing bene discontent with _Albertus Durerus_, his Geometricall Institutions in Dutch: or with _Gulielmus Xylander_, his learned translation of the first sixe bookes of _Euclide_, out of the Greke into the high Dutch. Nor with _Gualterus H. Riffius_, his Geometricall Volume: very diligently translated into the high Dutch tounge, and published. Nor yet the _Vniuersities_ of Spaine, or Portugall, thinke their reputation to be decayed: or suppose any their Studies to be hindred by the Excellent _P. Nonnius_, his Mathematicall workes, in vulgare speche by him put forth. Haue you not, likewise, in the French tounge, the whole Mathematicall Quadriuie? and yet neither Paris, Orleance, or any of the other Vniuersities of Fraunce, at any time, with the Translaters, or Publishers offended: or any mans Studie thereby hindred?

[3.]

And surely, the Common and Vulgar Scholer (much more, the Gramarian) before his comming to the _Vniuersitie_, shall (or may) be, now (according to _Plato_ his Counsell) sufficiently instructed in _Arithmetike_ and _Geometrie_, for the better and easier learning of all maner of _Philosophie_, _Academicall_, or _Peripateticall_. And by that meanes, goe more cherefully, more skilfully, and spedily forwarde, in his Studies, there to be learned. And, so, in lesse time, profite more, then (otherwise) he should, or could do.

[4.]

Also many good and pregnant Englishe wittes, of young Gentlemen, and of other, who neuer intend to meddle with the profound search and Studie of Philosophie (in the _Vniuersities_ to be learned) may neuerthelesse, now, with more ease and libertie, haue good occasion, vertuously to occupie the sharpnesse of their wittes: where, els (perchance) otherwise, they would in fond exercises, spend (or rather leese) their time: neither seruing God: nor furdering the Weale, common or priuate.

[5.]

And great Comfort, with good hope, may the _Vniuersities_ haue, by reason of this _Englishe_ +Geometrie, and Mathematicall Præface+, that they (hereafter) shall be the more regarded, esteemed, and resorted vnto. For, when it shall be knowen and reported, that of the _Mathematicall Sciences_ onely, such great Commodities are ensuing (as I haue specified): and that in dede, some of you vnlatined Studentes, can be good witnesse, of such rare fruite by you enioyed (thereby): as either, before this, was not heard of: or els, not so fully credited: “Well, may all men coniecture, that farre greater ayde, and better furniture, to winne to the Perfection of all Philosophie,

[Vniuersities.]

may in the Vniuersities be had: being the Storehouses & Threasory of all Sciences,

[□]

and all Artes, necessary for the best, and most noble State of Common Wealthes.”

[6.]

Besides this, how many a Common Artificer, is there, in these Realmes of England and Ireland, that dealeth with Numbers, Rule, & Cumpasse: Who, with their owne Skill and experience, already had, will be hable (by these good helpes and informations) to finde out, and deuise, new workes, straunge Engines, and Instrumentes: for sundry purposes in the Common Wealth? or for priuate pleasure? and for the better maintayning of their owne estate? I will not (therefore) fight against myne owne shadowe. For, no man (I am sure) will open his mouth against this Enterprise. No mã (I say) who either hath Charitie toward his brother (and would be glad of his furtherance in vertuous knowledge): or that hath any care & zeale for the bettering of the Cõmon state of this Realme. Neither any, that make accompt, what the wiser sort of men (Sage and Stayed) do thinke of them. To none (therefore) will I make any _Apologie,_ for a vertuous acte doing: and for cõmending, or setting forth,

Profitable Artes to English men, in the English toung. “But, vnto God our Creator, let vs all be thankefull: for that, +_As he, of his Goodnes, by his Powre, and in his wisedome,

[□]

hath Created all thynges, in Number, Waight, and Measure_+: So, to vs, of hys great Mercy, he hath reuealed Meanes, whereby, to atteyne the sufficient and necessary knowledge of the foresayd hys three principall Instrumentes: Which Meanes, I haue abundantly proued vnto you, to be the _Sciences_ and _Artes Mathematicall_.”

And though I haue ben pinched with straightnes of tyme: that, no way, I could so pen downe the matter (in my Mynde) as I determined: hopyng of conuenient laysure: Yet. if vertuous zeale, and honest Intent prouoke and bryng you to the readyng and examinyng of this Compendious treatise, I do not doute, but, as the veritie therof (accordyng to our purpose) will be euident vnto you: So the pith and force therof, will persuade you: and the wonderfull frute therof, highly pleasure you. And that you may the easier perceiue, and better remember, the principall pointes, whereof my Preface treateth,

[The Ground platt of this Præface in a Table.]

I will giue you the +Groundplatt+ of my whole discourse, in a Table annexed: from the first to the last, somewhat Methodically contriued.

If Hast, hath caused my poore pen, any where, to stumble: You will, (I am sure) in part of recompence, (for my earnest and sincere good will to pleasure you), Consider the rockish huge mountaines, and the perilous vnbeaten wayes, which (both night and day, for the while) it hath toyled and labored through, to bryng you this good Newes, and Comfortable profe, of Vertues frute.

So, I Commit you vnto Gods Mercyfull direction, for the rest: hartely besechyng hym, to prosper your Studyes, and honest Intentes: to his Glory, & the Commodity of our Countrey. _Amen_.

Written at my poore House At Mortlake.

Anno. 1570. February. 9.

[Decoration]

[Transcriber’s Note:

The “Groundplat” was printed in the form of a stemma, or tree, on an oversized fold-out page. The layout was impossible to reproduce for this e-text, so the information has been rearranged in nested-list form. Size markings (see note at beginning of e-text) are relative within each paragraph.]

J. DEE

+‡Here haue you (according to my promisse) the Groundplat of‡+ +my MATHEMATICALL Præface: annexed to _Euclide_ (now first)+ published in our Englishe tounge. An. 1570. Febr. 3.

+‡Sciences, and Artes Mathematicall,‡ are, either+

+‡Principall,‡ which are two, onely,+

+Arithmetike.+

+‡Simple‡+, Which dealeth with Numbers onely: and demonstrateth all their properties and appertenances: where, an Vnit, is Indiuisible. +‡Mixt‡+, Which with aide of Geometrie principall, demonstrateth some Arithmeticall Conclusion, or Purpose.

+Geometrie.+

+‡Simple‡+, Which dealeth with Magnitudes, onely: and demonstrateth all their properties, passions, and appertenances: whose Point, is Indiuisible. +‡Mixt‡+, Which with aide of Arithmetike principall, demonstrateth some Geometricall purpose, as +EVCLIDES ELEMENTES+.

+‡The vse‡ whereof, is either,+

In thinges Supernaturall, æternall, & Diuine: By Application, _Ascending_. In thinges Mathematicall: without farther Application. In thinges Naturall: both Substãtiall, & Accidentall, Visible, & Inuisible. &c. By Application: _Descending_.

The like Vses and Applications are, (though in a degree lower) in the +Artes Mathematicall Deriuatiue+.

+‡Deriuatiue‡ frõ the Principalls: of which, some haue+

+‡The names of‡ the Principalls: as,+

+_Arithmetike_, vulgar: which considereth+

--Arithmetike of most vsuall whole numbers: And of Fractions to them appertaining. --Arithmetike of Proportions. --Arithmetike Circular. --Arithmetike of Radicall Nũbers: Simple, Compound, Mixt: And of their Fractions. --Arithmetike of Cossike Nũbers: with their Fractions: And the great Arte of Algiebar.

+_Geometrie_, vulgar: which teacheth Measuring+

+‡At hand‡+

All Lengthes.--+Mecometrie.+ All Plaines: As, Land, Borde, Glasse, &c.--+Embadometrie.+ All Solids: As, Timber, Stone, Vessels, &c.--+Stereometrie.+

+‡With distãce‡+ from the thing Measured, as,

+‡How farre‡+, from the Measurer, any thing is: of him sene, on Land or Water: called +Apomecometrie+. +‡How high or deepe‡+, from the leuell of the Measurers standing, any thing is: Seene of hym, on Land or Water: called +Hypsometrie+. +‡How broad‡+, a thing is, which is in the Measurers view: so it be situated on Land or Water: called +Platometrie+.

+‡Of which‡ are growen the Feates & Artes of+

+Geodesie+: more cunningly to Measure and Suruey Landes, Woods, Waters. &c. +Geographie.+ +Chorographie.+ +Hydrographie.+ +Stratarithmetrie.+

+‡Propre names‡ as+,

+Perspectiue,+--Which demonstrateth the maners and properties of all Radiations: Directe, Broken, and Reflected.

+Astronomie,+--Which demonstrateth the Distances, Magnitudes, and all Naturall motions, Apparences, and Passions, proper to the Planets and fixed Starres: for any time, past, present, and to come: in respecte of a certaine Horizon, or without respecte of any Horizon.

+Musike,+--Which demonstrateth by reason, and teacheth by sense, perfectly to iudge and order the diuersitie of Soundes, hie or low.

+Cosmographie,+--Which, wholy and perfectly maketh description of the Heauenlym and also Elementall part of the World: and of these partes, maketh homologall application, and mutuall collation necessary.

+Astrologie,+--Which reasonably demonstrateth the operations and effectes of the naturall beames of light, and secrete Influence of the Planets, and fixed Starres, in euery Element and Elementall body: at all times, in any Horizon assigned.

+Statike,+--Which demonstrateth the causes of heauines and lightnes of all thinges: and of the motions and properties to heauines and lightnes belonging.

+Anthropographie,+ Which describeth the Nũber, Measure, Waight, Figure, Situation, and colour of euery diuers thing contained in the perfecte body of MAN: and geueth certaine knowledge of the Figure, Symmetrie, Waight, Characterization, & due Locall motion of any percell of the said body assigned: and of numbers to the said percell appertaining.

+Trochilike,+--Which demonstrateth the properties of all Circular motions: Simple and Compound.

+Helicosophie,+--Which demonstrateth the designing of all Spirall lines: in Plaine, on Cylinder, Cone, Sphære, Conoïd, and Sphæroid: and their properties.

+Pneumatithmie,+--Which demonstrateth by close hollow Geometricall figures (Regular and Irregular) the straunge properties (in motion or stay) of the Water, Ayre, Smoke, and Fire, in their Continuitie, and as they are ioyned to the Elementes next them.

+Menadrie,+--Which demonstrateth, how, aboue Natures Vertue, and power simple: Vertue and force, may be multiplied: and so to directe, to lift, to pull to, and to put or cast fro, any multiplied, or simple determined Vertue, Waight, or Force: naturally, not, so, directible, or moueable.

+Hypogeiodie,+--Which demonstrateth, how, vnder the Sphæricall Superficies of the Earth, at any depth, to any perpendicular line assigned (whose distance from the

perpendicular of the entrance: and the Azimuth likewise, in respecte of the sayd entrance, is knowen) certaine way, may be prescribed and gone, &c.

+Hydragogie,+--Which demonstrateth the possible leading of water by Natures law, and by artificiall helpe, from any head (being Spring, standing, or running water) to any other place assigned.

+Horometrie,+--Which demonstrateth, how, at all times appointed, the precise, vsuall denomination of time, may be knowen, for any place assigned.

+Zographie,+--Which demonstrateth and teacheth, how, the Intersection of all visuall Pyramids, made by any plaine assigned (the Center, distance, and lightes being determined) may be, by lines, and proper colours represented.

+Architecture,+--Which is a Science garnished with many doctrines, and diuers Instructions: by whose iudgement, all workes by other workmen finished, are iudged.

+Nauigation,+--Which demonstrateth, how, by the Shortest good way, by the aptest direction, and in the shortest time: a sufficient Shippe, betwene any two places (in passage nauigable) assigned, may be conducted: and in all stormes and naturall disturbances chauncing, how to vse the best possible meanes, to recouer the place first assigned.

+Thaumaturgike,+--Which geueth certaine order to make straunge workes, of the sense to be perceiued: and of men greatly to be wondred at.

+Archemastrie,+--Which teacheth to bring to actuall experience sensible, all worthy conclusions, by all the Artes Mathematicall purposed: and by true Naturall philosophie, concluded: And both addeth to them a farder Scope, in the termes of the same Artes: and also, by his proper Method, and in peculiar termes, procedeth, with helpe of the forsayd Artes, to the performance of complete Experiences: which, of no particular Arte, are hable (Formally) to be challenged.

+¶ Imprinted by _Iohn Day_.+

An. 1570. Feb. 25.

* * * * * * * * * * * * * *

Errors and Anomalies:

Unless otherwise noted, spelling and punctuation are unchanged. Errors are listed below, with the original form, if changed, shown in [brackets]. Unusual words include “fatch” (probably used as a variant of “fetch”) and the mathematical terms “sexagene” and “sexagesme”.

How, worldly goods: how, worldly dignitie [_“o” in second “worldly” invisible_] his most diligent hearers (so infinitely mought [hearers) so] the boundes, and duety of an Hydrographer [Hydographer] of the Grekes it is called _Eteromekes_ [_text unchanged: correct form is “Heteromekes”_] τὸ ὀτὶ [_accent unchanged_] in our worldly affaires [wordly] fall to worke.□. [_Some text readers may not display the oversized-asterisk symbol._] _Emptying the first._ [Emptyting] Απὸ τάυτης τῆς ἡμέρας, περὶ παντὸς, Αρχιμήδει λέγοντι πιζευτεόν [ἡμήρας ... πιζευτέομ] of the

suddeyne [snddeyne] that the right and absolute way may be had [he had] Georgic I: [_The quoted segments, each ending in “&c.”, are 438-439; 451-457; 463-464._]

Additional Notes:

The Greek letter η (eta) was consistently printed as if it were the ou-ligature 8. The Latin “-que” was written as an abbreviation resembling “-q´;”. It is shown here as [que].

Mathematical symbols seen in the section accompanying the diagrams could not be reproduced. The following substitutions were made: --The curly “P” used for “Pounds” is shown as {P}. --The “potestas” symbol, used to represent “x” (the unknown), is shown as {x}. --All roots were expressed as the “root” sign √ combined with symbols for the power of 2 (doubled for power of 4, or fourth root) and 3. They are shown as $^{2}\surd$ $^{3}\surd$ $^{4}\surd$.

Euclid:

The following Propositions were identified by number.

6.12: (How) to find a fourth (line) proportional to three given straight lines.

11.34: In equal parallelepipedal solids the bases are reciprocally proportional to the heights; and those parallelepipedal solids in which the bases are reciprocally proportional to the heights are equal.

11.36: If three straight lines are proportional, then the parallelepipedal solid formed out of the three equals the parallelepipedal solid on the mean which is equilateral, but equiangular with the aforesaid solid.

12.1: Similar polygons inscribed in circles are to one another as the squares on their diameters.

12.2: Circles are to one another as the squares on their diameters.

12.18 (“last”): Spheres are to one another in triplicate ratio of their respective diameters.